Spectral Structure, Multiscale Window Entropies, and Multi-Exchange Energy Landscapes of Magic Squares

Kyle Elliott Mathewson
Faculty of Science, University of Alberta
kyle.mathewson@ualberta.ca

Abstract

Every magic square of order n is a zero-energy ground state of the line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \sum_{L\in\mathcal{L}_n} R_L(S)^2 = \mathbf{z}^\top L_n \mathbf{z}$ on the symmetric group S_{n^2}, where $\mathbf{z} \in \mathbb{R}^{n^2}$ is the centred height field $Z_{ij} = 2s_{ij} - (n^2+1)$ and $L_n = A_n^\top A_n$ is the $(2n+2)$-line incidence operator. While previous papers in this series established the convex-polytopal geometry of Magic Gems (Paper I), their bivariate polynomial moments and isotropic inertia tensors (Paper II), and their rigid-body rotational and static mechanics (Paper III), a central inverse question remained open: how does the combinatorial energy landscape $\mathcal{H}_{\text{line}} : S_{n^2} \to \mathbb{Z}_{\geq 0}$ organize the $n^2!$ permutations around the magic ground-state manifold, why do naive continuous kernel relaxations and 2- or 3-exchange local searches get trapped, and what multi-exchange neighborhoods overcome those traps? Here we address this question across four scales. First, using the Rook's-graph decomposition of L_n, we show (Proposition 2.1) that restricting L_n to the (n^2-1)-dimensional zero-mean hyperplane $\mathbf{1}^\perp$ decouples the constant mode and reduces the irrational diagonal-coupled eigenvalues $\lambda_\pm(n)$ to exact integers—$\{0^{[n^2-2n-1]}, (n-2)^{[1]}, n^{[2n-1]}\}$ for even $n \geq 4$ and $\{0^{[n^2-2n-1]}, (n-1)^{[2]}, n^{[2n-2]}\}$ for odd $n \geq 3$—with mean eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$ and random-permutation expectation $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$, while linear dipole and diagonal balance ($\langle X, Z\rangle = \langle Y, Z\rangle = R_{D_\pm} = 0$) leaves a $(2n-4)$-dimensional subspace of higher-degree polynomial moments unconstrained (Remark 2.2). Second, we evaluate multiscale sliding-window energies and Shannon entropies $\widetilde{H}_p(k)$ across all 880 Frénicle order-4 magic squares and constructive families at $n \in \{3,4,5,6,8\}$, showing that vanishing interior 2×2 quartet energy ($W^{\text{int}}_{2\times 2} = 0$) is equivalent on even-order magic squares to vanishing toroidal quartet energy ($W^{\text{tor}}_{2\times 2} = 0$, Lemma 3.1), and proving the Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$ (Proposition 3.2), which classifies all 56 octupole zeros at order 4 (48 associative squares in Group III plus 8 coprime-resonance squares in Groups IX and X). Third, we perform an exhaustive census of all $9! = 362{,}880$ permutations of S_9 (Theorem 4.2), proving that $n = 3$ possesses *exactly* 8 *spurious strict* 2*-exchange local minima* forming a single dihedral orbit $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$ that traps 25.47% of S_9 under both 2-cycles and 3-cycles, yet escapes across an intermediate single-swap saddle ($\mathcal{H}^{\ddagger}_{\text{line}} = 12, \Delta\mathcal{H}^{\ddagger}_{\text{line}} = +4$) via a *unique* row-preserving $(2,2)$ double swap $(6\leftrightarrow 7)(3\leftrightarrow 4)$ (1 of 378 disjoint double swaps) that restores the deterministic ground-state basin of S_9 to 100.00%. Fourth, we show that continuous 2D Gaussian-kernel drift fails (0.0%) under both greedy cell snapping and exact $O(N^3)$ Hungarian bipartite projection due to convex amplitude averaging, whereas a discrete *Multi-Exchange Iterated Local Search (ME-ILS)* combining 2-exchange descent, line-preserving $(2,2)$ plaquette moves, candidate $(2,2)$ and row/column swaps, and thresholded basin-hopping achieves 100.0% ground-state convergence across all orders $n = 3, 4, \ldots, 10$ (up to $100! \approx 9.33 \times 10^{157}$ states), with restart counts declining above $n = 5$ as the diagonal-free $(2,2)$ subspace fraction $f_{\text{null}}(n) = 1 - 2/(n-1)^2$ approaches unity.

1 Introduction

A magic square of order n is a bijection $S : \{1,\ldots,n\}^2 \to \{1,\ldots,n^2\}$ whose n rows, n columns, and two main diagonals each sum to the magic constant $M_n = \frac{1}{2}n(n^2+1)$ [11, 8, 10, 1]. When

lifted into three-dimensional Euclidean space via centred integer coordinates

$$X_j = 2j - (n+1), \qquad Y_i = (n+1) - 2i, \qquad Z_{ij} = 2s_{ij} - (n^2+1), \tag{1}$$

each permutation $S \in S_{n^2}$ defines an n^2-point configuration $\mathcal{P}(S) = \{(X_j, Y_i, Z_{ij})\} \subset \mathbb{Z}^3$ whose convex hull $K(S) = \operatorname{conv}\mathcal{P}(S)$ is a *Magic Gem*. Across the first three papers of this series, we investigated the forward map from magic squares to geometric and mechanical invariants. In Paper I [19], we classified the polyhedral facial structure, vertex counts V, surface areas, volumes, and discrete symmetry groups of Magic Gems across the 880 Frénicle order-4 squares and higher-order families. Building on that geometric census, Paper II [18] proved that semi-magic row and column balance forces the bivariate polynomial moments $\langle X^j, Z\rangle = \langle Y^j, Z\rangle = 0$ to vanish for all $j \geq 0$, thereby diagonalizing the 3×3 inertia and covariance tensors of every magic square and generating universal Prouhet–Tarry–Escott power-sum identities. In Paper III [17], we tested whether those algebraic symmetries survive in physical rigid-body mechanics, demonstrating that every equal-vertex-mass Magic Gem is an oblate symmetric top ($I_{xx} = I_{yy} < I_{zz}$) immune to intermediate-axis Dzhanibekov tumbling, whereas continuous bulk polyhedra break covariance isotropy yet preserve exact static stability across all 50,334 facets of the 880 order-4 gems.

Despite this characterization of the *ground states*, an inverse landscape question remained open: *how are the magic ground states organized within the ambient $n^2!$-state permutation space S_{n^2}, and what discrete or continuous relaxation mechanisms can navigate from an arbitrary disordered permutation to a magic square?* For the purely constructive task of producing a single magic square of order n, classical closed-form algorithms (such as de La Loubère's Siamese rule for odd n, diagonal masks for doubly-even $n = 4k$, and Conway's LUX method for singly-even $n = 4k+2$) require only $O(n^2)$ operations [8, 1, 2], and specialized constraint-based local search (CBLS) heuristics [7, 27] scale to $n \sim 100$ by targeting defective lines. Treating magic-square relaxation instead as an energy-landscape problem on S_{n^2} connects classical recreational combinatorics to the theory of frustrated lattice systems [22, 24, 3], potential energy landscapes [29, 28], and stochastic local search [14]. In an interactive companion laboratory (`physicalSolver` [20]), we prototyped four representations of this landscape: a spectral decomposition of line residuals, a multiscale sliding-window entropy analyzer, a discrete multi-exchange local search engine, and a continuous 2D particle visualization.

Following Whitesides's outline-driven methodology for linking mathematical structure to quantitative computational experiments [31], this fourth paper formalizes the operators underlying `physicalSolver` and establishes exact algebraic identities and exhaustive computational censuses for each stage of the hierarchy. Let $\mathcal{L}_n$ denote the set of $2n+2$ magic lines (n rows, n columns, and 2 main diagonals), and let $R_L(S) = \sum_{(i,j)\in L} Z_{ij}$ be the centred residual of line $L \in \mathcal{L}_n$. Collecting the $2n+2$ line indicator vectors as rows of the incidence matrix $A_n \in \{0,1\}^{(2n+2)\times n^2}$, the *line-indicator Hamiltonian* takes the quadratic form

$$\mathcal{H}_{\text{line}}(S) = \sum_{L\in\mathcal{L}_n} R_L(S)^2 = \|A_n\mathbf{z}\|_2^2 = \mathbf{z}^\top L_n \mathbf{z}, \qquad L_n = A_n^\top A_n \in \mathbb{Z}^{n^2\times n^2}. \tag{2}$$

By construction, $\mathcal{H}_{\text{line}}(S) \geq 0$ for all $S \in S_{n^2}$, and $\mathcal{H}_{\text{line}}(S) = 0$ if and only if S is a magic square.

Summary of Main Results

Our investigation proceeds from the linear-algebraic spectrum of L_n on $\mathbf{1}^\perp$ to local multiscale sub-windows, discrete swap-graph basins on S_{n^2}, and multi-exchange local search across orders $n = 3\ldots 10$ (Table 1):

We begin in Section 2 by reviewing the spectral decomposition of $L_n = L_{\text{rc}} + D$ via the Rook's graph $K_n \square K_n$ [12, 4, 16] and characterizing the insufficiency of low-order spatial moments. In Proposition 2.1, we show that while the unprojected operator L_n on $\mathbb{R}^{n^2}$ exhibits two irrational eigenvalues $\lambda_\pm(n)$ caused by the overlap between $\mathbf{1}$ and the two main diagonals, restricting L_n to the zero-mean hyperplane $\mathbf{1}^\perp$ via $\widetilde{L}_n = P_{\mathbf{1}^\perp} L_n P_{\mathbf{1}^\perp}$ confines the rank-2 diagonal perturbation

Table 1: Overview of the four-scale landscape hierarchy and principal results established in Paper D.

Scale / Section	Operator / Observable	Formal Result / Census	Key Quantitative Invariant
Sec. 2: Zero-Mean Spectrum on $\mathbf{1}^\perp$	Line operator $L_n = A_n^\top A_n$ Random ensemble $\mathbb{E}[\mathcal{H}_{\text{line}}]$ Dipole + diag energy E_{low}	Proposition 2.1 ($\mathbf{1}^\perp$ spectrum & mean) Proposition 2.1 ($\bar{\mu}(\widetilde{L}_n) = 2$) Remark 2.2 ($2n-4$ poly. modes)	$\text{spec}(\widetilde{L}_n) \subset \mathbb{Z}$, $\dim\ker = n^2 - 2n - 1$ $\mathbb{E}[\mathcal{H}_{\text{line}}] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$ S_{low}: $E_{\text{low}} = 0$, $\mathcal{H}_{\text{line}} = 1{,}584$
Sec. 3: Multiscale Windows & Parity	2×2 quartet energy $W_{2\times 2}$ Shell quadrupoles $Q_{\text{corner}}, Q_{\text{edge}}$ Window entropy $\widetilde{H}_p(k)$	Lemma 3.1 (interior $\Leftrightarrow$ toroidal) Proposition 3.2 ($M_{xyz} = -8Q_{\text{corner}} - 3Q_{\text{edge}}$) Constructive census ($n = 3\ldots 8$)	$W^{\text{int}}_{2\times 2} = 0 \iff$ Group I (48/880) $M_{xyz} = 0$ on 48 Grp III +8 Grp IX/X Max $\widetilde{H}_1(2) = 0.99967$ (#123, Grp VI)
Sec. 4: Swap Landscape on S_{n^2}	Transposition cost $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$ S_9 exhaustive basin census S_{16} descent (2,000 starts)	Lemma 4.1 (rank-1 QAP update) Theorem 4.2 (8 traps $S^{(3)}_{\text{trap}}$, $\Delta\mathcal{H}^\ddagger_{\text{line}} = +4$) Order-4 local-minimum spectrum	$\Delta\mathcal{H}_{\text{line}} = -2d(g_a - g_b) + c_{ab}d^2$ ($c_{ab} \le 7$) 74.53% $\to$ 100.00% w/ unique $(2,2)$ 18.25% (2-cycle) $\to$ 25.45% w/ $(2,2)$
Sec. 5: Kernel Drift vs. Multi-Exchange	Continuous 2D kernel flow Multi-Exchange ILS (ME-ILS)	Regimes I–IV (greedy & Hungarian) 6-tier ablation & multi-order ($n = 3\ldots 10$)	$\mathcal{H}_{\text{ind}}$: 1681.9 $\to$ 1050.6 (0% discrete) 100.0% synthesis across all $n = 3\ldots 10$

inside $\ker(L_{\text{rc}})$ and yields purely integer eigenvalues: $\{0^{[n^2-2n-1]}, (n-2)^{[1]}, n^{[2n-1]}\}$ for even $n \ge 4$ and $\{0^{[n^2-2n-1]}, (n-1)^{[2]}, n^{[2n-2]}\}$ for odd $n \ge 3$. Both the trace identity $\text{tr}(\widetilde{L}_n) = 2(n^2-1)$ and finite-population sampling without replacement immediately give a mean zero-mean eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$ and random-permutation expectation $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$ (Figure 1a). Furthermore, Remark 2.2 notes that enforcing Paper II's linear dipole orthogonality $\langle X, Z\rangle = \langle Y, Z\rangle = 0$ together with diagonal balance $R_{D_+} = R_{D_-} = 0$ constrains only a 4-dimensional subspace of the $2n$-dimensional active image $\text{im}(\widetilde{L}_n) \subset \mathbf{1}^\perp$, leaving a $(2n-4)$-dimensional subspace of higher-degree polynomial row and column moments unconstrained, exemplified at $n = 4$ by an explicit permutation S_{low} with $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584$ (Figure 1b).

To resolve spatial structure inside the magic ground-state manifold $\ker(\widetilde{L}_n)$ where all 1D line residuals vanish, Section 3 evaluates sliding $k \times k$ sub-window energies and normalized Shannon entropies $\widetilde{H}_p(k)$ across constructive families ($n = 3, 4, 5, 6, 8$) and all 880 order-4 squares (Figure 2). After recording in Lemma 3.1 that vanishing of the $(2m-1)^2$ interior 2×2 quartet sums on any even-order magic square ($n = 2m$) forces all boundary-wrapping toroidal 2×2 quartets to vanish (isolating the 48 panmagic squares of Dudeney Group I at $n = 4$), we establish the Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$ in Proposition 3.2. This identity explains why the antipodal complement-break count $B_{\text{comp}}(S) \in \{0, 8\}$ is strictly binary across all 880 squares and proves that the mixed octupole moment $M_{xyz}(S)$ vanishes on all 48 Group III associative squares together with exactly 8 coprime-resonance squares in Groups IX and X (Figure 3).

Turning from static invariants to discrete relaxation dynamics on S_{n^2}, Section 4 applies the rank-1 transposition difference formula $\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) = -2d(g_a - g_b) + c_{ab}d^2$ ($c_{ab} \in \{2, \ldots, 7\}$, Lemma 4.1) across all $9! = 362{,}880$ permutations of S_9 (Theorem 4.2 and Figure 4). We prove that besides the 8 Lo Shu ground states ($\mathcal{H}_{\text{line}} = 0$), S_9 contains *exactly* 8 *spurious strict local minima* forming a single D_4 orbit $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$ ($E_{\text{std}} = 2$). On $S^{(3)}_{\text{trap}}$, every single 2-cycle ($\min\Delta\mathcal{H}_{\text{line}} = +4$) and every single 3-cycle ($\min\Delta\mathcal{H}_{\text{line}} = +8$) strictly increases $\mathcal{H}_{\text{line}}$. However, testing all $\frac{1}{2}\binom{9}{2}\binom{7}{2} = 378$ disjoint $(2,2)$ double transpositions proves that *one and only one*—the row-preserving double swap $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$, which traverses an intermediate single-swap saddle at $\mathcal{H}^\ddagger_{\text{line}} = 12$ ($\Delta\mathcal{H}^\ddagger_{\text{line}} = +4$)—escapes $S^{(3)}_{\text{trap}}$ directly to the Lo Shu ground state, expanding the deterministic ground-state basin of S_9 from 270,442/362,880 (74.53%) to 362,880/362,880 (100.00%). At order $n = 4$ (S_{16}), line-preserving $(2,2)$ double swaps similarly escape 16.57% of trapped 2-cycle local minima (4.83× more than 3-cycles at 3.43%), raising deterministic steepest-descent convergence from 18.25% to 25.45% (Figure 5).

Finally, Section 5 compares continuous 2D spatial kernel drift against discrete multi-exchange local search (Figure 6 and Table 7). We first demonstrate that naive 2D Gaussian-kernel particle relaxation fails (0.0% magic squares under both greedy cell snapping and exact $O(N^3)$ Hungarian bipartite matching) because the soft kernel surrogate $\mathcal{H}_{\text{ind}}(t)$ lowers energy via convex amplitude averaging in the interior of the assignment polytope rather than crossing discrete permutation saddles. By contrast, our discrete *Multi-Exchange Iterated Local Search (ME-ILS)* algorithm—combining $O(1)$ incremental 2-exchange steepest descent, whole-row/column permu-

tations, row/column-preserving $(2,2)$ plaquette moves, candidate-list $(2,2)$ swaps, and threshold-bounded basin-hopping restarts—achieves 100.0% ground-state synthesis across all orders $n = 3,4,5,6,7,8,9,10$ (Table 8, up to $100! \approx 9.33 \times 10^{157}$ states) and shows that mean restarts peak at $n = 5$ (8.03) before declining across $n = 6 \ldots 10$ (3.34 at $n = 10$) as the diagonal-free $(2,2)$ subspace fraction $f_{\text{null}}(n) = 1 - 2/(n-1)^2$ approaches unity (Figure 7).

2 Zero-Mean Spectrum of the Line-Incidence Operator and Moment Insufficiency

To analyze the global geometry of the line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_n \mathbf{z}$ defined in Equation 2, we first review the spectral decomposition of the line-incidence Gram operator $L_n = A_n^\top A_n$ on the ambient Euclidean space $\mathbb{R}^{n^2}$ and on the zero-mean hyperplane $\mathbf{1}^\perp = \{\mathbf{z} \in \mathbb{R}^{n^2} : \mathbf{1}^\top \mathbf{z} = 0\}$ where every centred permutation vector $\mathbf{z}(S)$ resides.

2.1 Rook's-Graph Incidence Structure and Zero-Mean Spectrum on $\mathbf{1}^\perp$

Let $\mathbf{r}_1, \ldots, \mathbf{r}_n \in \{0,1\}^{n^2}$ denote the row-indicator vectors of the $n \times n$ grid, $\mathbf{c}_1, \ldots, \mathbf{c}_n \in \{0,1\}^{n^2}$ the column-indicator vectors, and $\mathbf{d}_+, \mathbf{d}_- \in \{0,1\}^{n^2}$ the main and anti-diagonal indicator vectors. Then

$$L_n = A_n^\top A_n = L_{\text{rc}} + D, \qquad L_{\text{rc}} = \sum_{i=1}^{n} \mathbf{r}_i \mathbf{r}_i^\top + \sum_{j=1}^{n} \mathbf{c}_j \mathbf{c}_j^\top, \qquad D = \mathbf{d}_+ \mathbf{d}_+^\top + \mathbf{d}_- \mathbf{d}_-^\top. \tag{3}$$

In algebraic graph theory, the row-and-column incidence operator $L_{\text{rc}} = A_{\text{Rook}} + 2I_{n^2}$ is the signless incidence operator of the $n \times n$ Rook's graph $K_n \square K_n = (K_n \otimes I_n) \oplus (I_n \otimes K_n)$ [12, 4]. Because the n rows partition the n^2 cells and the n columns also partition the n^2 cells, $\sum_{i=1}^{n} \mathbf{r}_i = \sum_{j=1}^{n} \mathbf{c}_j = \mathbf{1}_{n^2}$, which reduces the rank of L_{rc} to $2n-1$. Since the two main diagonals $\mathbf{d}_+$ and $\mathbf{d}_-$ are linearly independent of the row and column span for all $n \geq 3$, $\operatorname{rank}(A_n) = 2n+1$, and the dimension of the magic nullspace $\ker(L_n) = \ker(A_n)$ is $\dim \ker(L_n) = n^2 - 2n - 1$, a classical vector-space invariant of magic squares [30, 15, 16]. On $\mathbb{R}^{n^2}$, the non-zero inner product $\mathbf{1}^\top \mathbf{d}_\pm = n$ mixes the constant mode $\mathbf{1}$ with the symmetric diagonal sum $\mathbf{s}_d = \mathbf{d}_+ + \mathbf{d}_-$, producing two irrational eigenvalues $\lambda_\pm(n)$. Projecting onto $\mathbf{1}^\perp$ decouples the constant vector and confines the rank-2 diagonal perturbation D entirely inside the $(n-1)^2$-dimensional kernel of the Rook's-graph operator L_{rc} (Figure 1a and Table 2):

Proposition 2.1 (Zero-Mean Spectrum and Random-Permutation Expectation of L_n)**.** *For every order $n \geq 3$, the line-incidence matrix $A_n \in \{0,1\}^{(2n+2)\times n^2}$ has rank $2n+1$, and its magic nullspace $\ker(L_n) = \ker(A_n) \subset \mathbf{1}^\perp$ has dimension*

$$\dim \ker(L_n) = n^2 - 2n - 1 = (n-1)^2 - 2. \tag{4}$$

On the ambient space $\mathbb{R}^{n^2}$, the $2n+1$ non-zero eigenvalues of L_n consist of two irrational roots $\lambda_\pm(n)$ together with integer eigenvalues:

$$\operatorname{spec}(L_n) = \begin{cases} \left\{0^{[n^2-2n-1]},\ \lambda_-(n)^{[1]},\ n^{[2n-1]},\ \lambda_+(n)^{[1]}\right\}, & \text{even } n \geq 4, \\ \left\{0^{[n^2-2n-1]},\ \lambda_-(n)^{[1]},\ (n-1)^{[1]},\ n^{[2n-2]},\ \lambda_+(n)^{[1]}\right\}, & \text{odd } n \geq 3, \end{cases} \tag{5}$$

where $\lambda_\pm(n) = \frac{1}{2}\left(3n \pm \sqrt{n(n+16)}\right)$ for even n and $\lambda_\pm(n) = \frac{1}{2}\left(3n+1 \pm \sqrt{n^2+14n+1}\right)$ for odd n. Let $P_{\mathbf{1}^\perp} = I_{n^2} - \frac{1}{n^2}\mathbf{1}\mathbf{1}^\top$ denote the orthogonal projector onto $\mathbf{1}^\perp$, and let $\widetilde{L}_n = P_{\mathbf{1}^\perp} L_n P_{\mathbf{1}^\perp}\big|_{\mathbf{1}^\perp}$ be the restricted operator on $\mathbf{1}^\perp$. On $\mathbf{1}^\perp$, all eigenvalues are integers:

$$\operatorname{spec}\left(\widetilde{L}_n\big|_{\mathbf{1}^\perp}\right) = \begin{cases} \left\{0^{[n^2-2n-1]},\ (n-2)^{[1]},\ n^{[2n-1]}\right\}, & \text{if } n \geq 4 \text{ is even}, \\ \left\{0^{[n^2-2n-1]},\ (n-1)^{[2]},\ n^{[2n-2]}\right\}, & \text{if } n \geq 3 \text{ is odd}. \end{cases} \tag{6}$$

Table 2: Spectral invariants of the unprojected operator L_n on $\mathbb{R}^{n^2}$ and zero-mean operator $\widetilde{L}_n$ on $\mathbf{1}^\perp$ across orders $n = 3 \dots 8$, verified by `verify_paper_d.py`.

n	$N = n^2$	$\operatorname{rank}(A_n)$	$\dim\ker(L_n)$	Irrational $\lambda_\pm(L_n)$	Zero-Mean $\operatorname{spec}(\widetilde{L}_n\|_{\mathbf{1}^\perp})$	$\operatorname{tr}(\widetilde{L}_n)$	$\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2$
3	9	7	2	1.3944, 8.6056	$\{0^{[2]}, 2^{[2]}, 3^{[4]}\}$	16	480
4	16	9	7	1.5279, 10.4721	$\{0^{[7]}, 2^{[1]}, 4^{[7]}\}$	30	2,720
5	25	11	14	3.1010, 12.8990	$\{0^{[14]}, 4^{[2]}, 5^{[8]}\}$	48	10,400
6	36	13	23	3.2554, 14.7446	$\{0^{[23]}, 4^{[1]}, 6^{[11]}\}$	70	31,080
7	49	15	34	4.9172, 17.0828	$\{0^{[34]}, 6^{[2]}, 7^{[12]}\}$	96	78,400
8	64	17	47	5.0718, 18.9282	$\{0^{[47]}, 6^{[1]}, 8^{[15]}\}$	126	174,720

Moreover, for every integer $n \geq 3$ (both odd and even), $\operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp}) = 2(n^2-1)$, so the mean eigenvalue of $\widetilde{L}_n$ on $\mathbf{1}^\perp$ is $\bar{\mu}(\widetilde{L}_n) = \operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp})/(n^2-1) = 2$, and the expected line-indicator energy of a uniform random permutation $S \sim \operatorname{Unif}(S_{n^2})$ with $\|\mathbf{z}\|^2 = \frac{1}{3}n^2(n^4-1)$ is

$$\mathbb{E}_{S\sim\operatorname{Unif}(S_{n^2})}\big[\mathcal{H}_{\text{line}}(S)\big] = 2\,\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1). \tag{7}$$

Proof. By standard spectral properties of the Rook's graph $K_n \square K_n$ [12, 4], L_{rc} decomposes $\mathbb{R}^{n^2}$ into three mutually orthogonal eigenspaces: the constant line $\operatorname{span}\{\mathbf{1}\}$ with eigenvalue $2n$ (multiplicity 1), the $(2n-2)$-dimensional zero-mean row and column space $\mathcal{V}_{\text{row}} \oplus \mathcal{V}_{\text{col}}$ with flat eigenvalue n (where $\mathcal{V}_{\text{row}} = \{\mathbf{u}\otimes\mathbf{1}_n : \mathbf{u}^\top\mathbf{1}_n = 0\}$ and $\mathcal{V}_{\text{col}} = \{\mathbf{1}_n\otimes\mathbf{v} : \mathbf{v}^\top\mathbf{1}_n = 0\}$), and the $(n-1)^2$-dimensional tensor-product kernel $\ker(L_{\text{rc}}) = \mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$ with eigenvalue 0.

Because every row and column intersects each main diagonal in exactly one cell, $\mathbf{r}_i^\top\mathbf{d}_\pm = \mathbf{c}_j^\top\mathbf{d}_\pm = 1$ for all i, j, so $P_{\mathbf{1}^\perp}\mathbf{d}_\pm = \mathbf{d}_\pm - \frac{1}{n}\mathbf{1}$ is orthogonal to $\mathcal{V}_{\text{row}} \oplus \mathcal{V}_{\text{col}}$ and lies entirely inside $\ker(L_{\text{rc}}) = \mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$. Thus $\mathcal{V}_{\text{row}} \oplus \mathcal{V}_{\text{col}} \subset \ker(D)$ remains an exact eigenspace of both L_n and $\widetilde{L}_n$ with eigenvalue n and multiplicity $2n-2$.

On the remaining 3-dimensional subspace $\operatorname{span}\{\mathbf{1}, \mathbf{d}_+, \mathbf{d}_-\}$, define $\mathbf{s}_d = \mathbf{d}_+ + \mathbf{d}_-$ and $\mathbf{a}_d = \mathbf{d}_+ - \mathbf{d}_-$. Since $\mathbf{a}_d \in \ker(L_{\text{rc}})$ and $\|\mathbf{a}_d\|^2 = 2n - 2\mathbf{d}_+^\top\mathbf{d}_-$, we have $D\mathbf{a}_d = n\mathbf{a}_d$ for even n ($\mathbf{d}_+^\top\mathbf{d}_- = 0$) and $D\mathbf{a}_d = (n-1)\mathbf{a}_d$ for odd n ($\mathbf{d}_+^\top\mathbf{d}_- = 1$ at the center cell). On $\operatorname{span}\{\mathbf{1}, \mathbf{s}_d\}$, L_n has the 2×2 representation $[L_n]_{\mathcal{B}} = \left[\begin{smallmatrix} 2n & 4 \\ n & n \end{smallmatrix}\right]$ for even n and $\left[\begin{smallmatrix} 2n & 4 \\ n & n+1 \end{smallmatrix}\right]$ for odd n, whose characteristic polynomials yield $\lambda_\pm(n)$ in Equation 5. Projecting onto $\mathbf{1}^\perp$ replaces $\mathbf{s}_d$ by $\widetilde{\mathbf{s}}_d = \mathbf{s}_d - \frac{2}{n}\mathbf{1} \in \ker(L_{\text{rc}})$, which has squared norm $\|\widetilde{\mathbf{s}}_d\|^2 = \|\mathbf{s}_d\|^2 - 4 = 2(n-2)$ for even n and $2(n-1)$ for odd n. Because $\widetilde{\mathbf{s}}_d$ and $\mathbf{a}_d$ are orthogonal vectors inside $\ker(L_{\text{rc}})$, restricting $D = \frac{1}{2}(\mathbf{s}_d\mathbf{s}_d^\top + \mathbf{a}_d\mathbf{a}_d^\top)$ to $\mathbf{1}^\perp$ yields eigenvalues $\frac{1}{2}\|\widetilde{\mathbf{s}}_d\|^2 = n-2$ and $\frac{1}{2}\|\mathbf{a}_d\|^2 = n$ for even n, or $n-1$ and $n-1$ for odd n, while their orthogonal complement in $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$ has dimension $(n-1)^2 - 2 = n^2 - 2n - 1$, proving Equation 4–Equation 6.

Finally, Equation 7 follows independently from both the operator trace and finite-population sampling without replacement. Spectrally, $\operatorname{tr}(L_n) = (2n+2)n = 2n(n+1)$ and $\hat{\mathbf{1}}^\top L_n \hat{\mathbf{1}} = 2n+2$, so $\operatorname{tr}(\widetilde{L}_n|_{\mathbf{1}^\perp}) = 2n(n+1) - 2(n+1) = 2(n^2-1)$ and $\bar{\mu}(\widetilde{L}_n) = 2$. Probabilistically, for a uniform random permutation of a zero-sum vector $\mathbf{z} \in \mathbf{1}^\perp$, any line $L \in \mathcal{L}_n$ of size n is a simple random sample without replacement from n^2 cells with finite-population correction factor $\frac{n^2-n}{n^2-1}$, giving $\mathbb{E}[R_L(S)^2] = n\big(\frac{n^2-n}{n^2-1}\big)\frac{\|\mathbf{z}\|^2}{n^2} = \frac{\|\mathbf{z}\|^2}{n+1}$ [9]. Summing across the $2n+2 = 2(n+1)$ lines immediately yields $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2(n+1)\frac{\|\mathbf{z}\|^2}{n+1} = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$. □

2.2 Insufficiency of Linear Dipole Moments and the $(2n-4)$-Dimensional Higher-Degree Moment Subspace

In Paper II [18] and Paper III [17], we proved that semi-magic row and column balance implies vanishing linear dipole moments $\langle X, Z\rangle = \langle Y, Z\rangle = 0$, so that every magic square has an isotropic inertia tensor. A natural question is whether the converse holds when supplemented by the two main diagonal constraints: if a permutation $S \in S_{n^2}$ satisfies zero dipole tilt ($\langle X, Z\rangle = \langle Y, Z\rangle = 0$) and zero diagonal residuals ($R_{D_+} = R_{D_-} = 0$), must S be a magic square? By Proposition 2.1, the answer is negative for all $n \geq 3$:

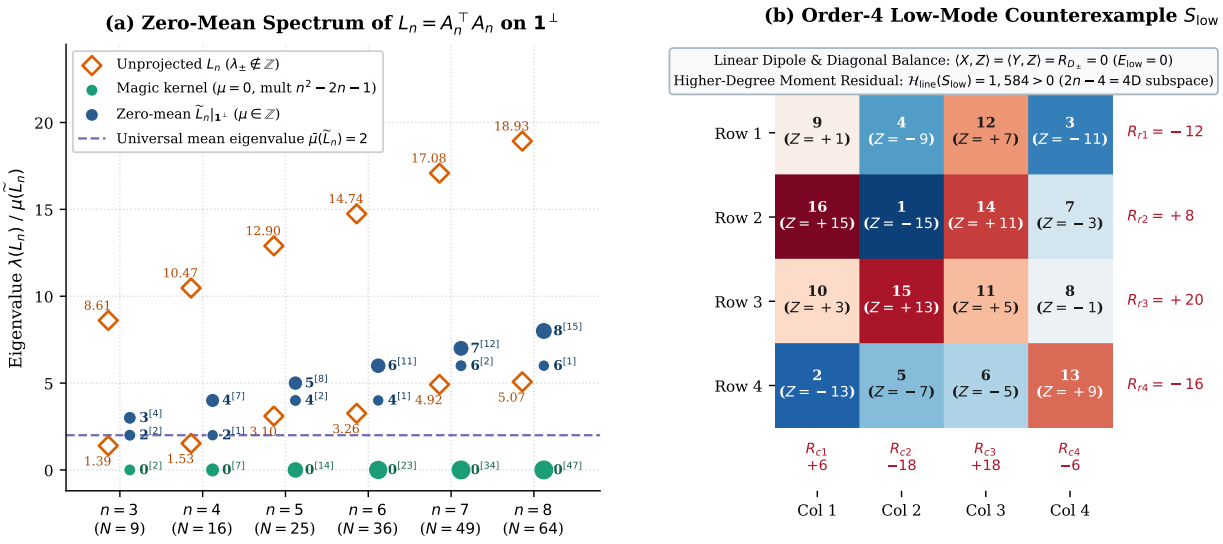


Figure 1: **Zero-mean spectrum of the line-incidence Laplacian $L_n = A_n^\top A_n$ and low-mode moment insufficiency.** **(a)** Comparison of the unprojected eigenvalues $\lambda(L_n)$ on $\mathbb{R}^{n^2}$ (hollow orange diamonds, showing the irrational branches $\lambda_\pm(n)$) against the zero-mean restricted spectrum $\mu(\widetilde{L}_n|_{\mathbf{1}^\perp})$ (filled navy and green circles, labeled $\mu^{[\text{mult}]}$) for orders $n = 3 \ldots 8$. Restricting to $\mathbf{1}^\perp$ yields integer eigenvalues with mean $\bar{\mu}(\widetilde{L}_n) = 2$ (dashed purple line). **(b)** Explicit 4×4 counterexample S_{low} (Remark 2.2) whose linear dipole moments $\langle X, Z\rangle = \langle Y, Z\rangle = 0$ and main diagonals $R_{D_\pm} = 0$ vanish ($E_{\text{low}} = 0$), yet whose quadratic and cubic row/column moments produce $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584 > 0$.

Remark 2.2 (Insufficiency of Linear Dipole and Diagonal Moments). Define the 4-mode dipole-plus-diagonal energy functional

$$E_{\text{low}}(S) = \langle X, Z\rangle^2 + \langle Y, Z\rangle^2 + R_{D_+}(S)^2 + R_{D_-}(S)^2. \tag{8}$$

Because $X \in \mathcal{V}_{\text{col}}$ and $Y \in \mathcal{V}_{\text{row}}$ are linear (degree-1) column and row coordinate vectors, they are mutually orthogonal and orthogonal to the projected diagonal vectors $\widetilde{\mathbf{d}}_\pm \in \mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$. Thus $\{X, Y, \widetilde{\mathbf{d}}_+, \widetilde{\mathbf{d}}_-\}$ spans a 4-dimensional subspace of the $2n$-dimensional active zero-mean image $\text{im}(\widetilde{L}_n) = \mathcal{V}_{\text{row}} \oplus \mathcal{V}_{\text{col}} \oplus \text{span}\{\widetilde{\mathbf{s}}_d, \mathbf{a}_d\} \subset \mathbf{1}^\perp$. Its orthogonal complement inside $\text{im}(\widetilde{L}_n)$ has dimension

$$\dim\big(\text{im}(\widetilde{L}_n) \cap \ker(E_{\text{low}})\big) = 2n - 4 = 2(n-2), \tag{9}$$

spanned by the $n-2$ higher-degree orthogonalized column polynomial moments $P_k(X) \in \mathcal{V}_{\text{col}}$ and $n-2$ row polynomial moments $P_k(Y) \in \mathcal{V}_{\text{row}}$ of degrees $2 \le k \le n-1$, all of which share the flat eigenvalue $\mu = n$ under $\widetilde{L}_n$. At order $n = 4$ ($2n - 4 = 4$ unconstrained polynomial modes corresponding to $\langle X^2, Z\rangle, \langle X^3, Z\rangle, \langle Y^2, Z\rangle, \langle Y^3, Z\rangle$), the explicit permutation

$$S_{\text{low}} = \begin{bmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{bmatrix} \in S_{16} \tag{10}$$

satisfies $\langle X, Z\rangle = 0$, $\langle Y, Z\rangle = 0$, $R_{D_+}(S_{\text{low}}) = 0$, and $R_{D_-}(S_{\text{low}}) = 0$ ($E_{\text{low}}(S_{\text{low}}) = 0$), yet exhibits non-zero row residuals $\mathbf{R}_{\text{row}} = (-12, +8, +20, -16)$, column residuals $\mathbf{R}_{\text{col}} = (+6, -18, +18, -6)$, and total line energy $\mathcal{H}_{\text{line}}(S_{\text{low}}) = 1{,}584 > 0$ (Figure 1b).

Remark 2.2 explains why macroscopic rigid-body torque and moment-of-inertia balance cannot synthesize magic squares on their own: any physical relaxation that couples only to global center-of-mass tilt ($\langle X, Z\rangle, \langle Y, Z\rangle$) and principal diagonal sums ($R_{D_\pm}$) constrains only 4 linear directions in $\mathbf{1}^\perp$ and is blind to the $2n-4$ higher-degree polynomial row and column moments ($\langle X^k, Z\rangle, \langle Y^k, Z\rangle$

for $2 \le k \le n-1$). At the same time, Proposition 2.1 defines the *nullspace-to-constraint ratio* on the zero-mean hyperplane $\mathbf{1}^{\perp}$:

$$\eta_{\mathrm{MC}}(n) = \frac{\dim\ker(\widetilde{L}_n)}{\operatorname{rank}(\widetilde{L}_n)} = \frac{n^2-2n-1}{2n}, \tag{11}$$

analogous to the ratio of internal zero-energy degrees of freedom to active linear constraints in Maxwell–Calladine constraint counting [21, 5]. Note an important distinction between continuous and discrete settings: in continuous linear algebra on $\mathbf{1}^{\perp}$, $\dim\ker(\widetilde{L}_n) = n^2-2n-1 > 0$ for every $n \ge 3$, so the linear system $A_n\mathbf{z} = \mathbf{0}$ always possesses a non-trivial nullspace. What makes small orders $n \in \{3,4,5\}$ combinatorially frustrated is the *discrete permutation constraint* $\mathbf{z} \in S_{n^2}$—that is, finding vertices of the permutahedron that lie inside $\ker(\widetilde{L}_n)$ when $\dim\ker(\widetilde{L}_n)$ is smaller than or comparable to the rank $2n$ ($\eta_{\mathrm{MC}}(3) = 0.33$, $\eta_{\mathrm{MC}}(4) = 0.88$, $\eta_{\mathrm{MC}}(5) = 1.40$). As $n \ge 6$ grows, the nullspace dimension expands quadratically while the constraint rank grows only linearly ($\eta_{\mathrm{MC}}(8) = 2.94$, $\eta_{\mathrm{MC}}(10) = 3.95$), producing the discrete dimensional relief quantified in Section 5.

3 Multiscale Window Entropies and 2×2 Quartet Equivalence

Once a permutation reaches the magic ground-state manifold $\ker(\widetilde{L}_n)$ where $\mathcal{H}_{\mathrm{line}}(S) = 0$, all $2n+2$ global 1D line sums are identical. However, distinct magic squares of the same order exhibit markedly different local 2D spatial organization [10, 23, 6]. To resolve this internal structure across scales, we evaluate sliding $k \times k$ sub-window energies and Shannon entropies exhaustively across all 880 Frénicle order-4 magic squares and higher-order constructive families.

3.1 Sliding-Window Shannon Entropy and Constructive Family Scaling

For any scale $k \in \{1,\dots,n\}$, an $n \times n$ grid contains $M_k = (n-k+1)^2$ interior contiguous $k \times k$ sub-windows $W_{r,c}^{(k)} = \{r,\dots,r+k-1\} \times \{c,\dots,c+k-1\}$ ($1 \le r,c \le n-k+1$). For any exponent $p \in \{1,2,4\}$, define the local ℓ_p^p window weight and normalized Shannon window entropy by

$$w_p\big(W_{r,c}^{(k)}\big) = \sum_{(i,j)\in W_{r,c}^{(k)}} |Z_{ij}|^p, \qquad p_{r,c}^{(k)} = \frac{w_p(W_{r,c}^{(k)})}{\sum_{r',c'} w_p(W_{r',c'}^{(k)})}, \qquad \widetilde{H}_p(k) = \frac{-\sum_{r,c} p_{r,c}^{(k)} \log_2 p_{r,c}^{(k)}}{\log_2\big((n-k+1)^2\big)}. \tag{12}$$

Here $\widetilde{H}_p(k) \in (0,1]$ measures the spatial uniformity of height magnitude across all $k \times k$ sub-windows ($\widetilde{H}_p(k) = 1$ if and only if every $k \times k$ window carries identical ℓ_p^p weight).

Figure 2a and Table 3 compare the multiscale entropy profile $\widetilde{H}_p(k)$ across five classical constructive families: odd-order de La Loubère (Siamese) squares ($n = 3, 5$) [8], doubly-even 4×4 diagonal-mask squares ($n = 4, 8$) [1], and Conway's LUX singly-even construction ($n = 6$) [2]. At $k = 2$, the 3×3 Lo Shu square achieves the highest local uniformity ($\widetilde{H}_1(2) = 0.99435$, $\widetilde{H}_2(2) = 0.99073$), whereas doubly-even mask constructions at $n = 4, 8$ concentrate extreme values symmetrically in corner and central blocks, lowering $\widetilde{H}_2(2)$ to 0.91498 ($n = 4$) and 0.90598 ($n = 8$). While the $n = 3, 4, 5, 8$ constructions are centrosymmetric ($B_{\mathrm{comp}} = 0$ complement breaks and $M_{xyz} = 0$), Conway's $n = 6$ LUX method breaks all 18/18 antipodal complement pairs and generates a non-zero mixed octupole moment $M_{xyz} = -132$.

3.2 Equivalence of Interior and Toroidal 2×2 Quartet Balance at Even Orders

Beyond magnitude entropy, the signed sum of each 2×2 quartet $Q_{r,c}(S) = Z_{r,c} + Z_{r+1,c} + Z_{r,c+1} + Z_{r+1,c+1}$ (with indices taken modulo n in the toroidal case) defines the *interior* and *toroidal* 2×2 *quartet energies*:

$$W_{2\times2}^{\mathrm{int}}(S) = \sum_{r=1}^{n-1}\sum_{c=1}^{n-1} Q_{r,c}(S)^2, \qquad W_{2\times2}^{\mathrm{tor}}(S) = \sum_{r=1}^{n}\sum_{c=1}^{n} Q_{r,c}(S)^2. \tag{13}$$

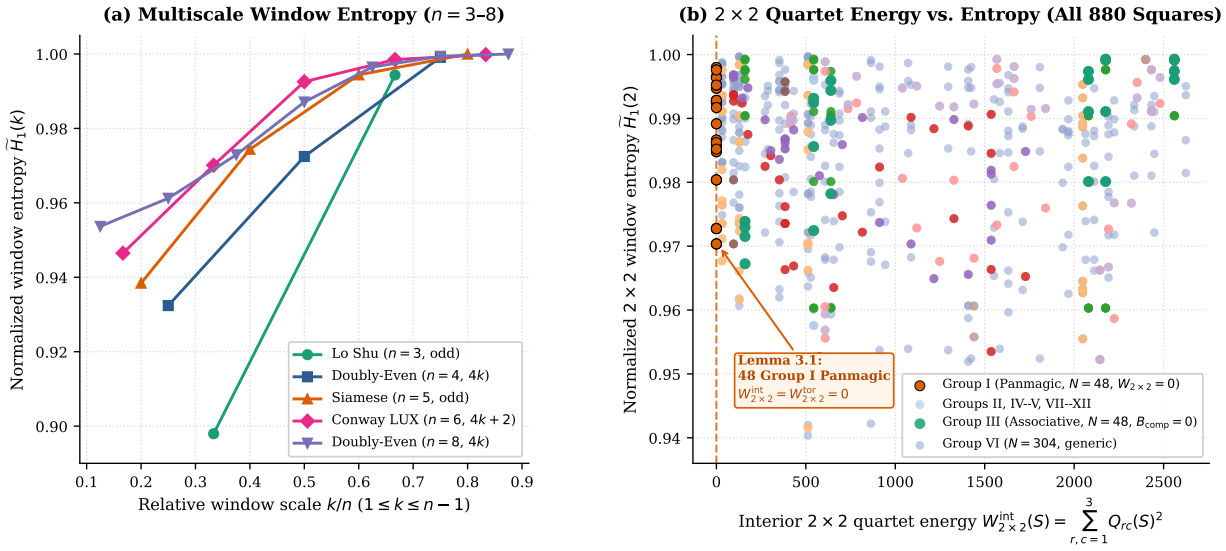


Figure 2: **Multiscale sliding-window Shannon entropy and 2×2 quartet energy census. (a)** Normalized ℓ_1 window entropy $\widetilde{H}_1(k)$ versus relative scale k/n across five constructive magic-square families ($n = 3$ Lo Shu, $n = 4, 8$ Doubly-Even, $n = 5$ Siamese, and $n = 6$ Conway LUX). **(b)** Joint distribution of interior 2×2 quartet residual energy $W^{\text{int}}_{2\times 2}(S)$ and window entropy $\widetilde{H}_1(2)$ across all 880 Frénicle order-4 magic squares partitioned by Dudeney Groups I–XII, illustrating Lemma 3.1: $W^{\text{int}}_{2\times 2}(S) = 0$ holds if and only if $W^{\text{tor}}_{2\times 2}(S) = 0$, isolating the 48 panmagic squares of Group I.

Table 3: Multiscale window entropy $\widetilde{H}_p(2)$, interior and toroidal 2×2 quartet energies $W_{2\times 2}$, complement breaks B_{comp}, and mixed octupole moment M_{xyz} across constructive magic square families ($n = 3 \ldots 8$).

Family	Order / Class	$\widetilde{H}_1(2)$	$\widetilde{H}_2(2)$	$W^{\text{int}}_{2\times 2}(S)$	$W^{\text{tor}}_{2\times 2}(S)$	Breaks B_{comp}	Octupole M_{xyz}
Lo Shu (Siamese)	$n = 3$ (odd)	0.99435	0.99073	80	240	0/4	0
Doubly-Even Mask	$n = 4$ $(4k)$	0.97245	0.91498	544	1,088	0/8	0
Siamese (de La Loubère)	$n = 5$ (odd)	0.97438	0.94254	16,320	25,200	0/12	0
Conway LUX Method	$n = 6$ $(4k + 2)$	0.97011	0.93161	84,640	97,440	18/18	−132
Doubly-Even Mask	$n = 8$ $(4k)$	0.96123	0.90598	112,320	149,760	0/32	0

At even orders $n = 2m$, an elementary row- and column-pair partition shows that interior quartet balance automatically forces toroidal boundary-wrapping quartet balance:

Lemma 3.1 (Interior–Toroidal 2×2 Quartet Equivalence)**.** *Let S be any magic square of even order $n = 2m \ge 4$. Then its $(2m-1)^2$ interior 2×2 blocks have zero centred sum ($W^{\text{int}}_{2\times 2}(S) = 0$)* if and only if *all $4m^2$ toroidal 2×2 blocks have zero centred sum ($W^{\text{tor}}_{2\times 2}(S) = 0$). In particular, across the* 880 *Frénicle magic squares of order $n = 4$, vanishing of the* 9 *interior quartets ($W^{\text{int}}_{2\times 2}(S) = 0$) is equivalent to vanishing of all* 16 *toroidal quartets ($W^{\text{tor}}_{2\times 2}(S) = 0$) and holds on* exactly the 48 panmagic (diabolic) squares of Dudeney Group I *(Figure 2b and Table 4).*

Proof. Clearly $W^{\text{tor}}_{2\times 2}(S) = 0 \implies W^{\text{int}}_{2\times 2}(S) = 0$. Conversely, suppose S is a magic square of even order $n = 2m$ ($\mathcal{H}_{\text{line}}(S) = 0$) with $Q_{r,c}(S) = 0$ for all interior blocks $1 \le r, c \le 2m-1$. For any horizontally wrapping 2×2 block $Q_{r,2m}(S)$ $(1 \le r \le 2m-1)$, the sum of rows r and $r+1$ vanishes by magic row balance and partitions into $m-1$ interior 2×2 blocks starting at even columns plus $Q_{r,2m}(S)$:

$$0 = R_{r_r}(S) + R_{r_{r+1}}(S) = \sum_{j=1}^{m-1} Q_{r,2j}(S) + Q_{r,2m}(S). \tag{14}$$

Since $Q_{r,2j}(S) = 0$ for $1 \le j \le m-1$, Equation 14 forces $Q_{r,2m}(S) = 0$. By the same argument on columns c and $c+1$, $Q_{2m,c}(S) = 0$ for $1 \le c \le 2m-1$, and finally $Q_{2m,2m}(S) = (R_{r_1}(S) + R_{r_{2m}}(S)) - \sum_{j=1}^{m-1} Q_{2m,2j}(S) = 0$. At $n = 4$ $(m = 2)$, $W^{\text{tor}}_{2\times 2}(S) = 0$ is a classical characterization of the 48 panmagic squares of Dudeney Group I [10, 23]. □

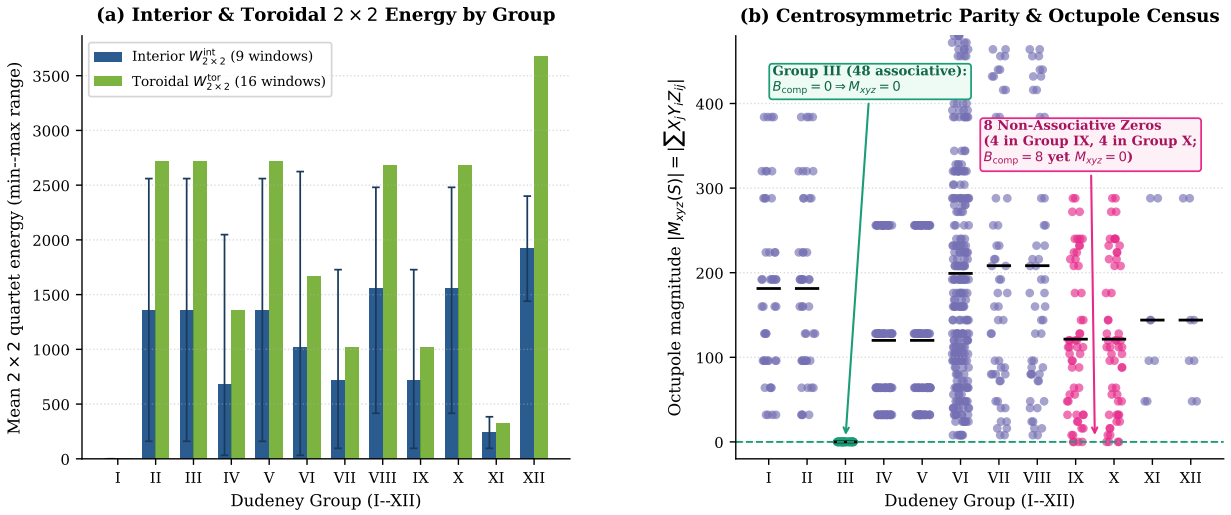


Figure 3: **Group-resolved 2×2 quartet energies and mixed octupole census across Dudeney Groups I–XII. (a)** Mean interior $W^{\text{int}}_{2\times2}$ (9 windows, navy bars with min–max range) and toroidal $W^{\text{tor}}_{2\times2}$ (16 windows, green bars) across the 12 Dudeney groups. **(b)** Magnitude of the mixed octupole moment $|M_{xyz}(S)| = |\sum X_j Y_i Z_{ij}|$ across all 880 squares (Proposition 3.2), highlighting the 48 associative squares of Group III ($B_{\text{comp}} = 0 \implies M_{xyz} = 0$) and the 8 anomalous non-associative octupole-zero squares in Groups IX and X ($B_{\text{comp}} = 8$ yet $M_{xyz} = 0$ via the Concentric Shell Quadrupole resonance $8Q_{\text{corner}} = -3Q_{\text{edge}}$).

3.3 Complement Parity and the Concentric Shell Quadrupole Identity

In Paper II [18] and Paper III [17], the mixed third-order moment (octupole) $M_{xyz}(S) = \sum_{i,j=1}^{n} X_j Y_i Z_{ij}$ governed the asymmetry between static and dynamic axes. To connect $M_{xyz}(S)$ to discrete cell pairing, define the *complement-break count* $B_{\text{comp}}(S)$ as the number of antipodal cell pairs $\{(i,j),(n+1-i,n+1-j)\}$ that fail the associative condition $Z_{ij} + Z_{n+1-i,n+1-j} = 0$ ($s_{ij} + s_{n+1-i,n+1-j} = n^2+1$). We now establish an exact Diophantine decomposition that explains both the symmetry-protected and accidental octupole zeros at order 4:

Proposition 3.2 (Concentric Shell Quadrupole Identity and Octupole Zero Classification)**.** *For any order-4 magic square S, partition the 16 cells into the 4 outer corners ($|X_j Y_i| = 9$), the 8 non-corner boundary edge cells ($|X_j Y_i| = 3$), and the 4 inner 2×2 cells ($|X_j Y_i| = 1$), with signed quadrupolar moments*

$$Q_{\text{corner}}(S) = (Z_{11} + Z_{44}) - (Z_{14} + Z_{41}), \qquad Q_{\text{inner}}(S) = (Z_{22} + Z_{33}) - (Z_{23} + Z_{32}),$$
$$Q_{\text{edge}}(S) = \big[(Z_{12} + Z_{43}) - (Z_{13} + Z_{42})\big] + \big[(Z_{21} + Z_{34}) - (Z_{31} + Z_{24})\big]. \tag{15}$$

Across all 880 Frénicle magic squares of order 4, the complement-break count $B_{\text{comp}}(S) \in \{0,1,\ldots,8\}$ takes only the two extreme values $\{0,8\}$*: $B_{\text{comp}}(S) = 0$ on the 48 associative squares of Dudeney Group III, and $B_{\text{comp}}(S) = 8$ on all remaining 832 squares, where the 8 complement pairs follow non-antipodal regular 2-factors of the 4×4 grid [10]. Moreover, diagonal balance $R_{D_+}(S) = R_{D_-}(S) = 0$ forces the exact concentric identity $Q_{\text{inner}}(S) = -Q_{\text{corner}}(S)$, reducing the mixed octupole moment of every 4×4 magic square to the Diophantine linear form*

$$M_{xyz}(S) = -9\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S) - Q_{\text{inner}}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S). \tag{16}$$

Consequently, $M_{xyz}(S) = 0$ holds if and only if $8\,Q_{\text{corner}}(S) = -3\,Q_{\text{edge}}(S)$, which is satisfied on exactly 56 of the 880 squares *(Figure 3b and Table 4). First, on the 48 associative squares of Group III, centrosymmetric parity $Z_{5-i,5-j} = -Z_{ij}$ forces $Q_{\text{corner}}(S) = Q_{\text{edge}}(S) = 0$ identically, yielding 48 symmetry-protected zeros. Second, across the 288 squares of Groups I, II, IV, and V, the complement pairings along rows, columns, or 2×2 blocks force $Q_{\text{edge}}(S) = 0$ while $Q_{\text{corner}}(S) \neq 0$,*

Table 4: Complete multiscale window and multipole census across the 12 Dudeney groups of the 880 order-4 magic squares, verified by `verify_paper_d.py`.

Dudeney Group	N_g	B_{comp}	Mean $\widetilde{H}_1(2)$	Mean $\widetilde{H}_2(2)$	$W^{\text{int}}_{2\times2}$ (mean [min, max])	$W^{\text{tor}}_{2\times2}$ (mean)	$\lvert M_{xyz}\rvert$ (mean, $\#=0$)
I (Panmagic)	48	8	0.98722	0.96742	0.0 [0, 0]	0.0	181.3 (0/48)
II (Semi-pan.)	48	8	0.98148	0.95260	1360.0 [160, 2560]	2720.0	181.3 (0/48)
III (Assoc.)	48	0	0.98807	0.96789	1360.0 [160, 2560]	2720.0	0.0 (48/48)
IV	96	8	0.97919	0.94745	680.0 [32, 2048]	1360.0	120.0 (0/96)
V	96	8	0.98831	0.96951	1360.0 [160, 2560]	2720.0	120.0 (0/96)
VI	304	8	0.98356	0.95870	1017.7 [32, 2624]	1671.2	199.2 (0/304)
VII	56	8	0.97874	0.94983	720.6 [96, 1728]	1015.4	208.3 (0/56)
VIII	56	8	0.98127	0.95081	1554.9 [416, 2480]	2684.0	208.3 (0/56)
IX	56	8	0.98432	0.95886	720.6 [96, 1728]	1015.4	121.4 (4/56)
X	56	8	0.98705	0.96844	1554.9 [416, 2480]	2684.0	121.4 (4/56)
XI	8	8	0.98516	0.96109	240.0 [96, 384]	320.0	144.0 (0/8)
XII	8	8	0.97871	0.94441	1920.0 [1440, 2400]	3680.0	144.0 (0/8)

so $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) \neq 0$ can never vanish in those groups. Third, the remaining 8 octupole zeros arise from coprime shell resonances among non-associative squares ($B_{\text{comp}} = 8$): specifically, 4 squares in Group IX (Frénicle ***#200, #208, #390, #711****) and 4 squares in Group X (****#492, #494, #794, #811****) satisfy $\big(Q_{\text{corner}}(S), Q_{\text{edge}}(S)\big) = (-3k, 8k)$ at $k = \pm4$ (****#208, #711, #492, #494****) and $k = -8$ (****#200, #390, #794, #811****), mapped bijectively between Groups IX and X by the simultaneous row/column parity permutation $\pi = (2, 4, 1, 3)$ and paired within each group by complementation $S \leftrightarrow 17\mathbf{1} - S$.*

Proof. Since $R_{D_+}(S) = (Z_{11} + Z_{44}) + (Z_{22} + Z_{33}) = 0$ and $R_{D_-}(S) = (Z_{14} + Z_{41}) + (Z_{23} + Z_{32}) = 0$, subtracting the two equations yields $Q_{\text{inner}}(S) = (Z_{22} + Z_{33}) - (Z_{23} + Z_{32}) = -(Z_{11} + Z_{44}) + (Z_{14} + Z_{41}) = -Q_{\text{corner}}(S)$. Expanding $M_{xyz}(S) = \sum_{i,j=1}^{4} X_j Y_i Z_{ij}$ with $(X_1, X_2, X_3, X_4) = (-3, -1, 1, 3)$ and $(Y_1, Y_2, Y_3, Y_4) = (3, 1, -1, -3)$ gives $M_{xyz}(S) = -9\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S) - Q_{\text{inner}}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$. Because $\gcd(8, 3) = 1$, vanishing of $M_{xyz}(S)$ requires $Q_{\text{corner}}(S) = -3k$ and $Q_{\text{edge}}(S) = 8k$ for some integer $k \in \mathbb{Z}$, verified exhaustively across all 880 squares in `verify_paper_d.py`. □

Across all 880 squares, the global maximum of 2×2 window entropy is achieved by Frénicle square `#123` in Group VI ($\widetilde{H}_1(2) = 0.999669$), while the global minimum occurs at square `#599` in Group VI ($\widetilde{H}_1(2) = 0.940345$). Notice also in Table 4 and Figure 3a that Groups II, III, and V share the exact same mean interior quartet energy $\langle W^{\text{int}}_{2\times2} \rangle = 1{,}360 = \|\mathbf{z}\|^2$ and mean toroidal quartet energy $\langle W^{\text{tor}}_{2\times2} \rangle = 2{,}720 = 2\|\mathbf{z}\|^2$.

4 Combinatorial Swap Landscape and Antipodal Traps

Whereas Section 2 showed that the continuous quadratic form $\mathbf{z}^\top L_n \mathbf{z}$ on $\mathbf{1}^\perp$ is convex and barrier-free, restricting $\mathbf{z}$ to the discrete permutation polytope S_{n^2} induces a rugged combinatorial energy landscape under elementary cell swaps [7, 27]. We now record the incremental transposition update for $\mathcal{H}_{\text{line}}(S)$ and perform an exhaustive basin census of S_9 ($n = 3$) alongside a large-scale relaxation census of S_{16} ($n = 4$).

4.1 Incremental Transposition Update

Let $\mathbf{g}(S) = \frac{1}{2}\nabla\mathcal{H}_{\text{line}}(\mathbf{z}) = L_n\mathbf{z} = A_n^\top \mathbf{R}(S) \in \mathbb{Z}^{n^2}$ denote the *cell-stress vector*, whose component $g_a(S) = \sum_{L \ni a} R_L(S)$ at cell $a \in \{1, \ldots, n^2\}$ is the sum of the centred residuals of the 2, 3, or 4 magic lines passing through cell a. Swapping the values at cells a and b via the 2-exchange transposition $\tau_{ab} \in S_{n^2}$ replaces $\mathbf{z}$ by $\mathbf{z}' = \mathbf{z} - d(\mathbf{e}_a - \mathbf{e}_b)$, where $d = Z_a - Z_b = 2(s_a - s_b)$. Expanding $\mathbf{z}'^\top L_n \mathbf{z}'$ yields the standard rank-1 quadratic assignment difference [27]:

Lemma 4.1 (Quadratic Transposition Difference Formula)**.** *For any permutation $S \in S_{n^2}$ and any pair of cells $a \neq b$ with height difference $d = Z_a - Z_b$, the change in the line-indicator Hamiltonian*

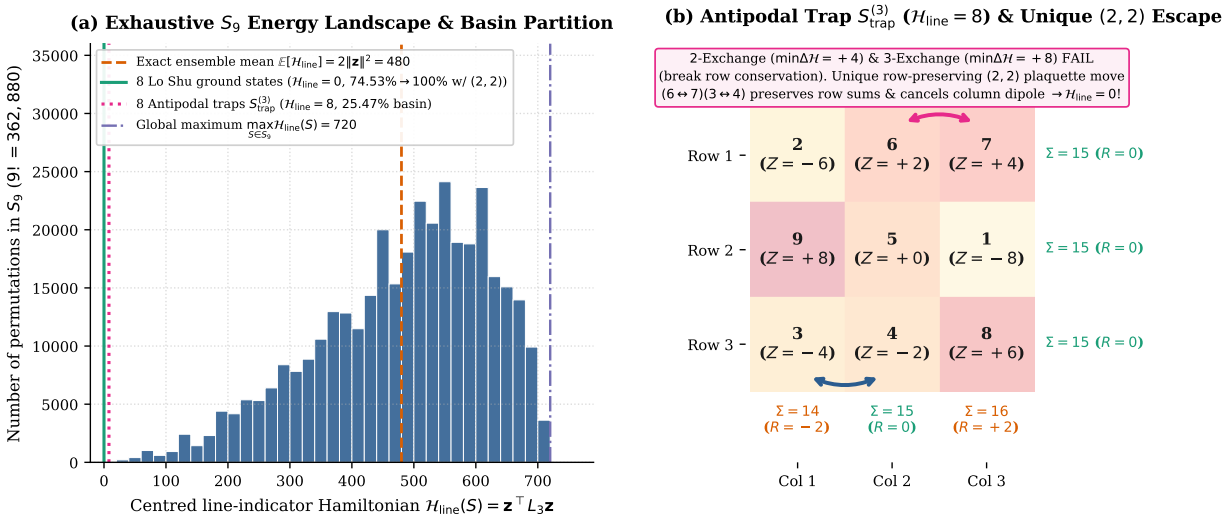


Figure 4: **Exhaustive S_9 permutation landscape (362,880 states) and the 8 antipodal traps at order $n = 3$. (a)** Exact histogram of the centred line Hamiltonian $\mathcal{H}_{\text{line}}(S)$ across all $9! = 362{,}880$ states of S_9, showing the ensemble mean $\mathbb{E}[\mathcal{H}_{\text{line}}] = 480$ (Proposition 2.1), the 8 Lo Shu ground states at $\mathcal{H}_{\text{line}} = 0$, and the 8 strict local-minimum traps $S^{(3)}_{\text{trap}}$ at $\mathcal{H}_{\text{line}} = 8$. **(b)** Structure of the canonical trap $S^{(3)}_{\text{trap}}$ (Theorem 4.2): all 3 rows, the middle column, and both main diagonals sum to 15 ($R = 0$), while the outer columns sum to 14 ($R = -2$) and 16 ($R = +2$). Every 2-cycle ($\min \Delta\mathcal{H}_{\text{line}} = +4$) and every 3-cycle ($\min \Delta\mathcal{H}_{\text{line}} = +8$) strictly increases energy; out of all 378 disjoint $(2, 2)$ double swaps, the *unique* escaping move is the row-preserving double swap $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$, which transitions directly to the Lo Shu ground state $\mathcal{H}_{\text{line}} = 0$ and restores the deterministic basin of S_9 to 100%.

under the transposition τ_{ab} is

$$\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) = \mathcal{H}_{\text{line}}(\tau_{ab}S) - \mathcal{H}_{\text{line}}(S) = -2d\big(g_a(S) - g_b(S)\big) + c_{ab}\, d^2, \tag{17}$$

where $c_{ab} = (L_n)_{aa} + (L_n)_{bb} - 2(L_n)_{ab} = \|A_n(\mathbf{e}_a - \mathbf{e}_b)\|_2^2 \in \{2, 3, 4, 5, 6, 7\}$ counts the number of magic lines in $\mathcal{L}_n$ that pass through exactly one *of the two distinct cells a and b ($c_{ab} \in \{3, 4, 5, 6\}$ for $n = 3$, $\{2, 3, 4, 5, 6\}$ for even $n \geq 4$, and $\{2, 3, 4, 5, 6, 7\}$ for odd $n \geq 5$, since only the unique central cell of odd n lies on all 4 lines). In particular, $\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) < 0$ if and only if $(g_a - g_b)/d > c_{ab}/2$.*

In the language of conserved-order-parameter lattice gases [13], Lemma 4.1 shows that a 2-exchange τ_{ab} is driven by the linear stress difference $-2d(g_a - g_b)$ but opposed by the quadratic self-energy barrier $c_{ab}d^2$ of the c_{ab} magic lines altered by the swap. When line residuals $R_L(S)$ are small (± 2 in centred units, or ± 1 in standard units), the linear stress release $-2d(g_a - g_b)$ has magnitude $O(|d|)$, whereas the self-energy penalty $c_{ab}d^2 \geq 8$ (since $|d| \geq 2$ and $c_{ab} \geq 2$) grows quadratically with the height jump $|d|$ and traps 2-exchange steepest descent in local minima.

4.2 Exhaustive S_9 Landscape Census and the 8 Antipodal Traps of Order 3

At order $n = 3$, the symmetric group S_9 has $9! = 362{,}880$ states, permitting an exact, exhaustive classification of the entire combinatorial energy landscape (Theorem 4.2 and Figure 4):

Theorem 4.2 (Exhaustive S_9 Landscape, 8 Strict Local Minima, and Unique $(2, 2)$ Escape)**.** *Across all* $9! = 362{,}880$ *permutations of S_9 under the line-indicator Hamiltonian $\mathcal{H}_{\text{line}}(S)$, there are* exactly 16 local minima *under the $\binom{9}{2} = 36$ elementary 2-cycle transpositions, all* 16 *of which are strict local minima ($\Delta\mathcal{H}_{\text{line}}(\tau_{ab}) > 0$ for all $a \neq b$). These* 16 *attractors partition into two dihedral orbits of size* 8*: the* 8 *global ground states ($\mathcal{H}_{\text{line}} = 0$) forming the dihedral orbit $D_4 \cdot S_{\text{LoShu}}$, and* exactly 8 spurious strict local minima *at $\mathcal{H}_{\text{line}} = 8$ ($E_{\text{std}} = 2$) forming the single dihedral D_4 orbit*

of the antipodal trap

$$S^{(3)}_{\text{trap}} = \begin{bmatrix} 2 & 6 & 7 \\ 9 & 5 & 1 \\ 3 & 4 & 8 \end{bmatrix}, \qquad \mathbf{Z}\big(S^{(3)}_{\text{trap}}\big) = \begin{bmatrix} -6 & +2 & +4 \\ +8 & 0 & -8 \\ -4 & -2 & +6 \end{bmatrix}, \qquad \mathcal{H}_{\text{line}}\big(S^{(3)}_{\text{trap}}\big) = 8. \tag{18}$$

In $S^{(3)}_{\text{trap}}$*, six of the eight magic lines (all three rows, the middle column, and both main diagonals) have zero residual* ($R_L = 0$)*, while the two outer columns have residuals* $R_{c_1} = -2$ ($\Sigma = 14$) *and* $R_{c_3} = +2$ ($\Sigma = 16$)*. Every single* 2*-cycle transposition strictly increases* $\mathcal{H}_{\text{line}}$ *by at least* $+4$ ($\min_{\tau_{ab}} \Delta\mathcal{H}_{\text{line}} = +4$)*, and* every single 3-cycle permutation among all $2\binom{9}{3} = 168$ directed 3-cycles also strictly increases $\mathcal{H}_{\text{line}}$ by at least $+8$. *Furthermore, across all* $\frac{1}{2}\binom{9}{2}\binom{7}{2} = 378$ *disjoint* $(2,2)$ *double transpositions on* $S^{(3)}_{\text{trap}}$*,* one and only one *strictly decreases* $\mathcal{H}_{\text{line}}$*: the row-preserving double swap* $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$ *at cells* $(1,2) \leftrightarrow (1,3)$ *and* $(3,1) \leftrightarrow (3,2)$*, which maps* $S^{(3)}_{\text{trap}}$ *directly to the Lo Shu ground state*

$$S_{\text{LoShu}} = \begin{bmatrix} 2 & 7 & 6 \\ 9 & 5 & 1 \\ 4 & 3 & 8 \end{bmatrix}, \qquad \mathcal{H}_{\text{line}}(S_{\text{LoShu}}) = 0. \tag{19}$$

Consequently, under deterministic 2*-cycle steepest descent (choosing the lexicographically first pair* (a,b) *minimizing* $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$*),* 270,442 *of the* 362,880 *states* (74.5266%) *converge to the* 8 *Lo Shu ground states, while the remaining* 92,438 *states* (25.4734%) *are captured by the* 8 *antipodal traps* $D_4 \cdot S^{(3)}_{\text{trap}}$*. Augmenting steepest descent with the* 18 *line-preserving* $(2,2)$ *double swaps whenever a* 2*-cycle local minimum is reached expands the deterministic ground-state basin to* 362,880/362,880 (100.00%)*, independent of the tie-breaking rule.*

Proof. All 362,880 permutations of S_9 (spanning $\mathcal{H}_{\text{line}}(S) \in [0, 720]$), their 36 transposition deltas $\Delta\mathcal{H}_{\text{line}}(\tau_{ab})$ via Equation 17, all 168 directed 3-cycle deltas, all 378 disjoint $(2,2)$ double-swap deltas on $S^{(3)}_{\text{trap}}$, and the steepest-descent attractor maps are evaluated exhaustively in `verify_paper_d.py` (Figure 4). To see algebraically why no local minima exist at $\mathcal{H}_{\text{line}} = 4$ ($E_{\text{std}} = 1$) and why $S^{(3)}_{\text{trap}}$ in Equation 18 has a unique escaping $(2,2)$ move: in standard units $s_{ij} - 5 \in \{-4, \ldots, 4\}$, $\sum_{i=1}^{n} R^{\text{std}}_{r_i} = \sum_{j=1}^{n} R^{\text{std}}_{c_j} = 0$, so a single row or column cannot have a lone non-zero residual of ± 1 without another row or column having a compensating residual, while any state with only $R^{\text{std}}_{D_\pm} = \pm 1$ is immediately relaxed by an off-diagonal 2-cycle. At $S^{(3)}_{\text{trap}}$, $E_{\text{std}}(S^{(3)}_{\text{trap}}) = (-1)^2 + (+1)^2 = 2$ ($\mathcal{H}_{\text{line}} = 8$). Because every row sum of $S^{(3)}_{\text{trap}}$ is already 15, any single 2-cycle between different rows alters two row sums by $\pm d \neq 0$ (adding $2d^2 \geq 2$ to the row energy), while any within-row 2-cycle changes a diagonal sum. Similarly, any 3-cycle either touches two cells in one row and one cell in another row (breaking both row sums) or touches one cell in each of the three rows (which cannot preserve all three row sums since the three permuted integers are distinct). By contrast, in the row-preserving $(2,2)$ double transposition $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$, performing the first intra-row transposition $\tau_1 = (6 \leftrightarrow 7)$ in Row 1 shifts (R_{c_2}, R_{c_3}) from $(0, +2)$ to $(+2, 0)$ and shifts R_{D_-} from 0 to -2, passing through an intermediate single-swap saddle state $\tau_1 S^{(3)}_{\text{trap}}$ at $\mathcal{H}^{\ddagger}_{\text{line}} = (-2)^2 + (+2)^2 + (-2)^2 = 12$ (an activation barrier of $\Delta\mathcal{H}^{\ddagger}_{\text{line}} = +4$ above $S^{(3)}_{\text{trap}}$), after which the conjugate intra-row transposition $\tau_2 = (3 \leftrightarrow 4)$ in Row 3 shifts (R_{c_1}, R_{c_2}) by $(+2, -2)$ and R_{D_-} by $+2$, annihilating all three residuals ($\Delta\mathcal{H}_2 = -12$) directly to $\mathcal{H}_{\text{line}}(S_{\text{LoShu}}) = 0$. □

Structurally, Theorem 4.2 illustrates why 4-element row- or column-preserving $(2,2)$ plaquette moves succeed where 2-cycles and 3-cycles fail, in close analogy to collective loop updates in ice-type models [24, 3]. In $S^{(3)}_{\text{trap}}$, all three rows already satisfy the zero-residual constraint $R_{r_i} = 0$. Any single 2-exchange between rows or any 3-exchange violates row conservation and creates non-zero row defects ($R_{r_i} \neq 0$). Only a row-preserving $(2,2)$ double swap—which acts within two rows simultaneously, passing through a virtual single-swap saddle at $\mathcal{H}^{\ddagger}_{\text{line}} = 12$ ($\Delta\mathcal{H}^{\ddagger}_{\text{line}} = +4$) via $\tau_1 = (6 \leftrightarrow 7)$ and cancelling the intermediate diagonal and middle-column residuals via $\tau_2 = (3 \leftrightarrow 4)$—preserves all row sums while eliminating the outer-column dipole.

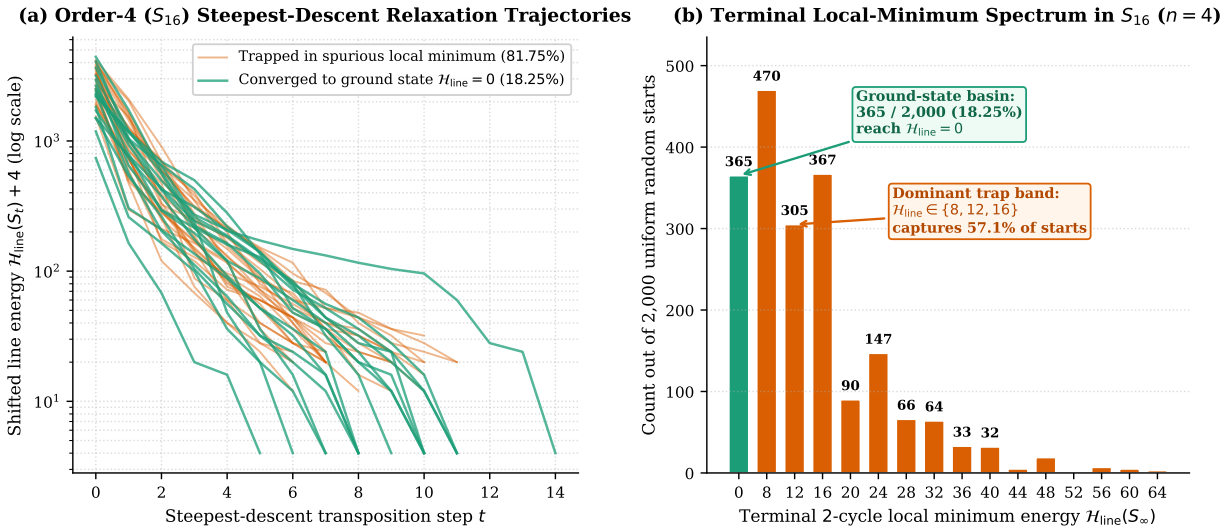


Figure 5: **Order-4 (S_{16}) steepest-descent relaxation trajectories and local-minimum spectrum (2,000 trials). (a)** Representative steepest-descent trajectories of $\mathcal{H}_{\text{line}}(S_t) + 4$ (log scale) versus transposition step t, contrasting trajectories that reach the magic ground state $\mathcal{H}_{\text{line}} = 0$ (green, 18.25%) against trajectories trapped in frustrated local minima (orange, 81.75%). **(b)** Histogram of terminal 2-cycle local-minimum energies $\mathcal{H}_{\text{line}}(S_\infty)$ across 2,000 uniform random starts in S_{16}. While 365/2,000 (18.25%) reach $\mathcal{H}_{\text{line}} = 0$, 57.1% of all starts become trapped in the low-energy frustration band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$.

Table 5: Summary of combinatorial swap-landscape invariants for S_9 ($n = 3$, exhaustive 362,880 states) and S_{16} ($n = 4$, 2,000 random starts), verified by `verify_paper_d.py`.

Landscape Observable / Metric	Order $n = 3$ (S_9, Exhaustive)	Order $n = 4$ (S_{16}, 2,000 Trials)
Total permutation state space $n^2!$	362,880	20,922,789,888,000
Magic ground states ($\mathcal{H}_{\text{line}} = 0$)	8 ($1 \times D_4$ orbit)	7,040 ($880 \times D_4$ orbits)
Ensemble mean energy $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2$	480	2,720
Elementary 2-cycle neighborhood size $\binom{n^2}{2}$	36	120
Directed 3-cycle neighborhood size $2\binom{n^2}{3}$	168	1,120
Line-preserving $(2,2)$ double-swap neighborhood $2\binom{n}{2}^3$	18	432
Spurious 2-cycle local minima ($\mathcal{H}_{\text{line}} > 0$)	Exactly 8 (at $\mathcal{H}_{\text{line}} = 8$)	Abundant ($\mathcal{H}_{\text{line}} \in \{8, 12, 16, \dots\}$)
Trapped minima escaped by a single 3-cycle	0/8 (0.00%)	56/1,635 (3.43%)
Trapped minima escaped by a line-preserving $(2,2)$ swap	8/8 (100.00%)	271/1,635 (16.57%)
2-cycle steepest-descent ground-state basin	270,442/362,880 (74.53%)	365/2,000 (18.25%)
2-cycle + line-preserving $(2,2)$ steepest-descent basin	362,880/362,880 (100.00%)	509/2,000 (25.45%)

4.3 Order-4 (S_{16}) Steepest-Descent and $(2,2)$-Escape Census

At order $n = 4$, the state space S_{16} grows to $16! \approx 2.09 \times 10^{13}$ permutations, while the number of distinct magic ground states is $880 \times 8 = 7{,}040$. There are $\binom{4}{2}\binom{4}{2}^2 = 216$ row-preserving and 216 column-preserving $(2,2)$ double swaps (432 total line-preserving $(2,2)$ moves). To quantify local-minimum trapping and escape mechanisms in S_{16}, we performed 2,000 independent steepest-descent trials starting from uniform random permutations (Figure 5 and Table 5).

As shown in Figure 5b and Table 5, pure 2-cycle steepest descent in S_{16} reaches a magic ground state ($\mathcal{H}_{\text{line}} = 0$) in 365/2,000 trials (18.25%), terminating in a mean of 8.18 steps at a mean residual energy $\langle \mathcal{H}_{\text{line}}(S_\infty) \rangle = 14.43$ (median 12.0). The trapped configurations cluster heavily in the lowest frustration band $\mathcal{H}_{\text{line}} \in \{8, 12, 16\}$, which accounts for $470 + 305 + 367 = 1{,}142/2{,}000$ (57.1%) of all random starts. Across the 1,635 trapped 2-cycle local minima, testing all $2\binom{16}{3} = 1{,}120$ directed 3-cycles escapes only 56/1,635 (3.43%). By contrast, testing the 432 line-preserving $(2,2)$ double swaps immediately escapes 271/1,635 (16.57%) of the trapped states—nearly 5× the escape rate of 3-cycles despite searching 2.6× fewer candidates (432 vs. 1,120)—and raises pure zero-temperature steepest-descent ground-state convergence from 18.25% to 25.45% (509/2,000).

Table 6: Parameters and trajectory invariants across the four continuous 2D spatial kernel regimes at order $n = 4$ (250 steps, $\Delta t = 0.025$), verified by `verify_paper_d.py`.

Dynamical Regime	(k_f, k_r, k_w, k_c)	(γ_m, k_s, T)	$\mathcal{H}_{\text{ind}}(0)$	$\mathcal{H}_{\text{ind}}(250)$	Min $\mathcal{H}_{\text{ind}}$	Mean Δr_{rms}	Max Δr_{rms}
I. Stiff Lattice	$(0.0, 0.0, 7.0, 0.0)$	$(0.94, 0.00, 0.0)$	1681.89	1681.89	1681.89	0.0000	0.0000
II. Weak Drift	$(1.0, 0.4, 2.6, 0.0)$	$(0.82, 0.00, 0.5)$	1681.89	1539.22	1539.22	0.0774	0.1134
III. Strong Field	$(3.2, 1.0, 0.8, 0.4)$	$(0.60, 0.00, 1.2)$	1681.89	1194.88	1194.88	0.2654	0.4646
IV. Target-Biased	$(2.2, 0.5, 1.8, 0.5)$	$(0.75, 0.85, 0.4)$	1681.89	1050.55	1050.55	0.4281	0.6381

5 Continuous Spatial Kernel Relaxation vs. Multi-Exchange Local Search

A natural question arising from the discrete local-minimum traps of Section 4 is whether relaxing the rigid permutation lattice into continuous two-dimensional space $\mathbb{R}^2$ allows particles to slide around combinatorial barriers. In our companion interactive laboratory (`physicalSolver` [20]), we implemented a 2D particle dynamics model (`bubbleTrinityApp.js`) that embeds the n^2 numbers as mobile particles in $\mathbb{R}^2$, alongside a discrete multi-exchange local search engine on S_{n^2}. Here we contrast why naive 2D spatial kernel interpolation fails (0.0% ground-state recovery even under exact Hungarian bipartite projection) against the discrete multi-exchange iterated local search mechanisms that achieve 100.0% ground-state convergence across orders $n = 3 \ldots 10$.

5.1 Why Naive 2D Spatial Kernel Relaxation Fails: Convex Amplitude Averaging vs. Permutation Saddles

Suppose the n^2 integers are represented by continuous coordinates $\mathbf{r}_a(t) = (x_a(t), y_a(t)) \in \mathbb{R}^2$ $(a = 1, \ldots, n^2)$ initialized at the grid centers $\mathbf{r}_a^{(0)} = (c_a - \frac{n+1}{2}, r_a - \frac{n+1}{2})$. At each lattice site $b \in \{1, \ldots, n^2\}$, soft Gaussian kernel interpolation defines the induced height vector $\widetilde{\mathbf{z}}(t)$ and surrogate line energy $\mathcal{H}_{\text{ind}}(t)$:

$$W_{ab}(t) = \frac{\exp\big(-2\|\mathbf{r}_a(t) - \mathbf{r}_b^{(0)}\|_2^2\big)}{\sum_{a'=1}^{n^2} \exp\big(-2\|\mathbf{r}_{a'}(t) - \mathbf{r}_b^{(0)}\|_2^2\big)}, \qquad \widetilde{\mathbf{z}}(t) = W(t)^\top \mathbf{z}, \qquad \mathcal{H}_{\text{ind}}(t) = \|A_n \widetilde{\mathbf{z}}(t)\|_2^2. \tag{20}$$

In the damped Langevin surrogate of `bubbleTrinityApp.js`, each particle evolves with step size $\Delta t = 0.025$ and velocity persistence $\gamma_m \in [0, 1)$ under the update

$$\begin{aligned} \mathbf{v}_a(t + \Delta t) &= \gamma_m \mathbf{v}_a(t) + \Delta t \Big[\mathbf{F}_a^{\text{well}} + \mathbf{F}_a^{\text{cont}} + \mathbf{F}_a^{\text{rep}} + \mathbf{F}_a^{\text{field}} + \sigma_T \boldsymbol{\xi}_a(t)\Big], \\ \mathbf{r}_a(t + \Delta t) &= \mathbf{r}_a(t) + \Delta t\, \mathbf{v}_a(t + \Delta t), \end{aligned} \tag{21}$$

combining harmonic lattice confinement $\mathbf{F}_a^{\text{well}} = -k_w(\mathbf{r}_a - \mathbf{r}_a^{(0)})$, radial containment $\mathbf{F}_a^{\text{cont}} = -k_c \mathbf{r}_a$, softened core repulsion $\mathbf{F}_a^{\text{rep}} = c_r k_r \sum_{b \neq a} (\mathbf{r}_a - \mathbf{r}_b)/(\|\mathbf{r}_a - \mathbf{r}_b\|^2 + \epsilon_0)^{3/2}$ $(c_r = 0.08, \epsilon_0 = 0.15)$, target-tracking drift $\mathbf{F}_a^{\text{field}} = (c_f k_f + c_s k_s)(\mathbf{r}_a^{\text{target}} - \mathbf{r}_a)$ $(c_f = 0.25, c_s = 0.65)$ toward the stress-weighted centroid $\mathbf{r}_a^{\text{target}}(t) = \sum_{b=1}^{n^2} \sigma_{ab}(t) \mathbf{r}_b^{(0)}$ with $\sigma_{ab}(t) \propto \exp(-Z_a(A_n^\top A_n \widetilde{\mathbf{z}}(t))_b / 250)$, and Gaussian thermal noise $\boldsymbol{\xi}_a(t) \sim \mathcal{N}(\mathbf{0}, I_2)$ scaled by $\sigma_T = 0.12\,T$. Note that because $\mathbf{F}_a^{\text{field}}$ tracks $\mathbf{r}_a^{\text{target}}(t)$ rather than the conservative gradient $-\nabla_{\mathbf{r}_a} \mathcal{H}_{\text{ind}}(t)$ and lacks a concave penalty forcing $W(t)$ toward permutation vertices, Equation 21 is an empirical heuristic visualization flow rather than a conservative Birkhoff-polytope solver.

Figure 6a and Table 6 evaluate Equation 21 across four parameter regimes starting from a 4-swap perturbation of Dürer's 4×4 square ($\mathcal{H}_{\text{ind}}(0) = 1{,}681.89$, discrete $\mathcal{H}_{\text{line}} = 2{,}336$). As the field coupling increases from Regime I ($\Delta r_{\text{rms}} = 0.0000$) and Regime II ($\langle \Delta r_{\text{rms}} \rangle = 0.0774$) to Regime III ($\max \Delta r_{\text{rms}} = 0.4646$) and Regime IV ($\max \Delta r_{\text{rms}} = 0.6381$), the soft kernel-interpolated energy $\mathcal{H}_{\text{ind}}(t)$ drops by 37.5% to 1,050.55. However, projecting the continuous particle positions $\{\mathbf{r}_a(t)\}$ back onto a discrete permutation $S \in S_{16}$ yields 0.0% magic squares across all four regimes (Tier 0 in Figure 6b). To verify that this failure is not merely an artifact of greedy nearest-cell assignment collisions ($\min_t \mathcal{H}_{\text{line}}(S_{\text{greedy}}) = 1{,}480$ in Regime IV and 2,336 in Regimes I–III),

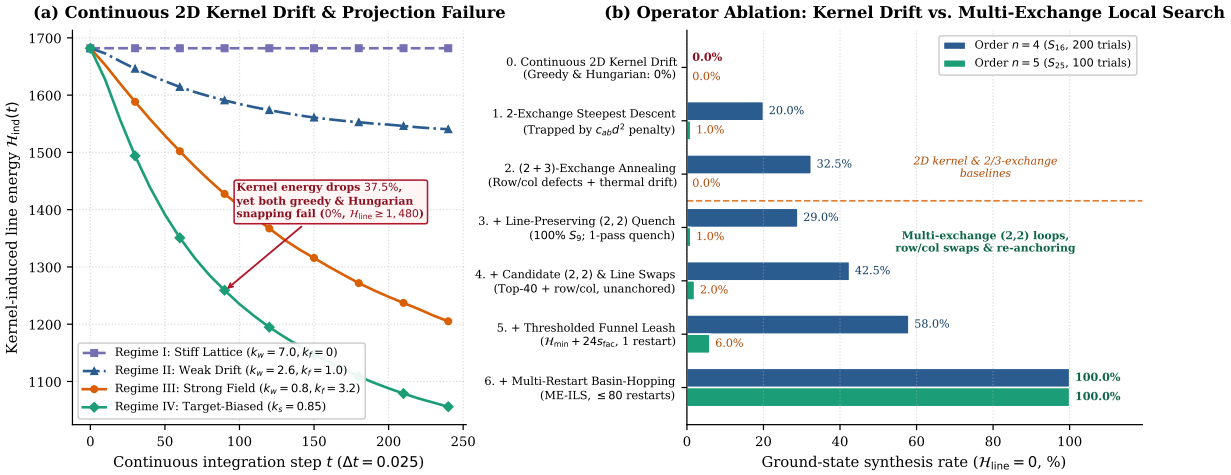


Figure 6: **Continuous 2D spatial kernel trajectories and discrete local-search mechanism ablation.** **(a)** Trajectories of the kernel-induced line energy $\mathcal{H}_{\text{ind}}(t)$ across 250 integration steps ($\Delta t = 0.025$) for the four parameter regimes of Table 6, starting from a 4-swap perturbation of Dürer's 4×4 square ($\mathcal{H}_{\text{ind}}(0) = 1{,}681.89$). Although continuous drift reduces $\mathcal{H}_{\text{ind}}(t)$ by 37.5%, both greedy snapping and exact Hungarian projection fail (0%, $\mathcal{H}_{\text{line}} \geq 1{,}480$) because $\mathcal{H}_{\text{ind}}(t)$ relaxes via convex amplitude averaging rather than discrete permutation swaps. **(b)** Operator ablation study (Table 7) contrasting continuous 2D drift and 2/3-exchange heuristics (Tiers 0–2) against line-preserving $(2,2)$ plaquette moves, candidate-list $(2,2)$ swaps with row/column permutations, thresholded funnel re-anchoring, and multi-restart iterated local search (Tiers 3–6) at orders $n = 4$ (200 trials) and $n = 5$ (100 trials).

we also evaluated exact $O(N^3)$ Hungarian / Kuhn–Munkres minimum-weight bipartite matching $\min_{\pi \in S_{n^2}} \sum_a \|\mathbf{r}_a(t) - \mathbf{r}^{(0)}_{\pi(a)}\|_2^2$ at every integration step. Even under exact Hungarian assignment, the discrete energy never improves beyond its initial value ($\min_t \mathcal{H}_{\text{line}}(S_{\text{Hung}}) = 2{,}336$ across all four regimes, 0.0% convergence). Because the Gaussian kernel matrix $W(t)$ in Equation 20 is column-stochastic rather than doubly stochastic and $\mathcal{H}_{\text{ind}}(t)$ is convex in $\widetilde{\mathbf{z}}$, particles lower $\mathcal{H}_{\text{ind}}(t)$ by clustering near intermediate cell boundaries to average out positive and negative heights in the interior of the assignment polytope, rather than executing the discrete multi-particle exchanges required to cross permutation saddles.

5.2 Multi-Exchange Iterated Local Search and Operator Ablation

To understand what discrete move neighborhoods are required to navigate the permutation landscape $\mathcal{H}_{\text{line}}(S)$ on S_{n^2}, it is important to distinguish two goals. For the pure constructive problem of generating a single magic square of order n, classical closed-form rules (Siamese, doubly-even mask, Conway LUX) require $O(n^2)$ operations, and specialized constraint-based local search (CBLS) / adaptive tabu heuristics [7, 27] reach $n \sim 100$ by restricting swaps to defective lines. Here, our goal is instead to map the *topological frustration of the energy landscape* $\mathcal{H}_{\text{line}}(S)$ under progressively richer permutation cycle neighborhoods. Table 7 and Figure 6b report a controlled ablation study across six discrete and continuous tiers at orders $n = 4$ (200 random starts) and $n = 5$ (100 random starts).

As established in Remark 2.2 and Section 5.1, macroscopic dipole/diagonal balance (blind to the $2n-4$ higher-degree polynomial moments) and continuous 2D kernel drift (Tier 0) both fail to reach $\mathcal{H}_{\text{line}} = 0$. On the discrete permutation graph, pure 2-exchange steepest descent (Tier 1, analogous to zero-temperature Kawasaki pair exchange [13]) halts whenever $-2d(g_a - g_b) + c_{ab}d^2 \geq 0$ for all pairs (a, b), succeeding on 74.5% of S_9, 20.0% of S_{16}, and only 1.0% of S_{25}. Adding 3-exchange moves under a 1,500-step simulated annealing schedule (Tier 2, [14]) fails to escape the order-3 trap $S^{(3)}_{\text{trap}}$ (0/8 escapes) and achieves only 32.5% at $n = 4$ and 0.0% at $n = 5$, because 3-cycles cannot swap elements between two columns while preserving their row sums, and unanchored thermal random walks drift out of narrow low-energy funnels.

Table 7: Controlled ablation study contrasting continuous 2D kernel drift and 2/3-exchange heuristics (Tiers 0–2) against cooperative $(2,2)$ plaquette moves, row/column permutations, thresholded funnel re-anchoring, and multi-restart basin-hopping (Tiers 3–6), certified by `verify_paper_d.py`.

Tier	Relaxation / Local-Search Mechanism	Move Neighborhood	Primary Bottleneck or Escape Mechanism	$n=3$ (S_9)	$n=4$ (S_{16})	$n=5$ (S_{25})
–	Linear Dipole & Diagonal Balance	4 global linear modes	Blind to $2n-4$ higher-degree polynomial moments (Remark 2.2: S_{low}, $\mathcal{H}_{\text{line}}=1{,}584$)	Fails	Fails	Fails
0	Continuous 2D Kernel Drift + Projection	Greedy & Hungarian snap	Convex kernel averaging in interior of assignment polytope ($\min \mathcal{H}_{\text{line}} \geq 1{,}480$)	0.0%	0.0%	0.0%
1	2-Exchange Steepest Descent [13]	$\binom{n^2}{2}$ pair swaps	Traps when quadratic penalty $c_{ab}d^2 \geq 8$ blocks $-2d(g_a-g_b)$ (Lemma 4.1)	74.5%	20.0%	1.0%
2	$(2+3)$-Exchange Simulated Annealing [14]	2- & 3-cycles ($B=1{,}500$)	3-cycles break row/col conservation (0/8 on $S^{(3)}_{\text{trap}}$); thermal walk loses funnel	74.5%	32.5%	0.0%
3	+ Line-Preserving $(2,2)$ Plaquette Quench	$2\binom{n}{2}^3$ row/col $(2,2)$ loops	Preserves row/col sums while cancelling conjugate column/row dipoles (1 quench)	**100.0**%	29.0%	1.0%
4	+ Candidate $(2,2)$ & Row/Col Permutations	Top-40 $(2,2)$ + $2\binom{n}{2}$ lines	Zero-barrier row/col swaps shear diagonals (5 unanchored kicks, 1 restart)	**100.0**%	42.5%	2.0%
5	+ Thresholded Funnel Re-Anchoring	Leash $\mathcal{H}_{\text{line}} \leq \mathcal{H}_{\min} + 24s_{\text{fac}}$	Snaps back to deepest minimum $S_{\min}$ on high kicks (18 kicks, 1 restart)	**100.0**%	58.0%	6.0%
6	+ Multi-Restart Basin-Hopping (ME-ILS) [29]	≤ 80 re-anchored restarts	Iterated local search across independent funnel basins (18 kicks/restart)	**100.0**%	**100.0**%	**100.0**%

Table 8: Multi-order synthesis benchmark of Multi-Exchange Iterated Local Search (ME-ILS) across orders $n = 3,4,5,6,7,8,9,10$ with order-scaled thresholds $s_{\text{fac}}(n) = \max(1, \lfloor (n/4)^2 \rfloor)$, certified by `verify_paper_d.py`. We report the nullspace-to-constraint ratio $\eta_{\text{MC}}(n) = (n^2-2n-1)/(2n)$, the fraction $f_{\text{null}}(n) = 1 - 2/(n-1)^2$ of the semi-magic $(2,2)$ subspace $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$ orthogonal to both diagonals, and the per-step 2-cycle pair neighborhood $\binom{n^2}{2}$.

Order n	State Space $n^2!$	Nullity $\dim\ker(\widetilde{L}_n)$	Ratio η_{MC}	Diag-Free $(2,2)$ f_{null}	Pairs $\binom{n^2}{2}$	Success ($\mathcal{H}_{\text{line}}=0$)	Mean Descents	Mean Pair Evals	Mean Restarts	Mean Time
$n=3$	$9! = 3.63\times10^5$	2	0.33	50.0%	36	**200/200 (100.0%)**	5.1	184	0.00	0.1 ms
$n=4$	$16! = 2.09\times10^{13}$	7	0.88	77.8%	120	**200/200 (100.0%)**	52.9	6,348	0.70	4.9 ms
$n=5$	$25! = 1.55\times10^{25}$	14	1.40	87.5%	300	**100/100 (100.0%)**	466.6	139,980	8.03	108.9 ms
$n=6$	$36! = 3.72\times10^{41}$	23	1.92	92.0%	630	**100/100 (100.0%)**	446.8	281,484	7.28	208.8 ms
$n=7$	$49! = 6.08\times10^{62}$	34	2.43	94.4%	1,176	**100/100 (100.0%)**	267.5	314,580	4.02	218.0 ms
$n=8$	$64! = 1.27\times10^{89}$	47	2.94	95.9%	2,016	**100/100 (100.0%)**	294.5	593,712	4.32	407.0 ms
$n=9$	$81! = 5.80\times10^{120}$	62	3.44	96.9%	3,240	**50/50 (100.0%)**	293.0	949,320	4.06	658.1 ms
$n=10$	$100! = 9.33\times10^{157}$	79	3.95	97.5%	4,950	**50/50 (100.0%)**	258.5	1,279,575	3.34	880.8 ms

Our *Multi-Exchange Iterated Local Search (ME-ILS)* algorithm overcomes these bottlenecks across Tiers 3–6. First, augmenting 2-exchange steepest descent with the $2\binom{n}{2}^3$ row- and column-preserving $(2,2)$ plaquette moves (Tier 3—analogous to divergence-free 4-body loop updates in ice-type lattice models [24, 3], bounded by single-swap saddle filter $\Delta\mathcal{H}_{\text{line}}(\tau_1) \leq 80 s_{\text{fac}}(n)$ where $s_{\text{fac}}(n) = \max(1, \lfloor (n/4)^2 \rfloor)$) eliminates all traps on S_9 (100.0%) and raises single-pass quench recovery on S_{16} from 20.0% to 29.0%. Second, expanding the local escape neighborhood at 2-exchange minima to include whole-row and whole-column permutations ($2\binom{n}{2}$ moves that preserve all row and column sums while altering only the two main diagonals) together with disjoint pairs selected from the top $K=40$ single swaps with $\Delta\mathcal{H}_{\text{line}}(\tau_1) \leq 48 s_{\text{fac}}(n)$ (Tier 4, with 5 unanchored basin kicks in a single restart) raises $n=4$ convergence to 42.5%. Third, introducing a threshold-bounded re-anchoring leash (Tier 5, which records the deepest local minimum $S_{\min}$ found within a restart and snaps back to $S_{\min}$ whenever a 2-swap kick followed by steepest descent exceeds $\mathcal{H}_{\text{line}}(S_{\min}) + 24 s_{\text{fac}}(n)$ across 18 kicks) prevents random walks from escaping the local funnel and raises single-restart $n=4$ convergence to 58.0%. Finally, wrapping this thresholded funnel search in an outer multi-restart basin-hopping loop (≤ 80 restarts, Tier 6 [29, 28]) achieves 100.0% ground-state convergence across all orders $n = 3,4,5,6,7,8,9,10$.

Table 8 and Figure 7 summarize the performance of ME-ILS across all eight orders $n = 3,4,5,6,7,8,9,10$ (900/900 total trials reaching $\mathcal{H}_{\text{line}}=0$, spanning up to $100! \approx 9.33\times10^{157}$ states at $n=10$). The data highlight both the geometric origin of the drop in restart counts at $n \geq 6$ and the polynomial scaling of per-step neighborhood evaluations. Every row- or column-preserving $(2,2)$ plaquette move acts via a vector $(\mathbf{e}_{r_1} - \mathbf{e}_{r_2}) \otimes (\mathbf{e}_{c_1} - \mathbf{e}_{c_2})$ inside the $(n-1)^2$-dimensional semi-magic subspace $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}} = \ker(L_{\text{rc}})$ (Proposition 2.1). Because the two main diagonals impose only 2 linear constraints $(\widetilde{\mathbf{s}}_d, \mathbf{a}_d)$ on $\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}}$, the fraction of the $(2,2)$ plaquette subspace that lies in the exact magic nullspace $\ker(\widetilde{L}_n)$ is

$$f_{\text{null}}(n) = \frac{\dim\ker(\widetilde{L}_n)}{\dim(\mathcal{V}_{\text{row}} \otimes \mathcal{V}_{\text{col}})} = \frac{n^2-2n-1}{(n-1)^2} = 1 - \frac{2}{(n-1)^2}, \tag{22}$$

alongside the nullspace-to-constraint ratio $\eta_{\text{MC}}(n) = (n^2-2n-1)/(2n)$ (Equation 11). At $n=5$ ($\eta_{\text{MC}} = 1.40$, $f_{\text{null}} = 87.5\%$), discrete permutation frustration peaks at 8.03 restarts (466.6 descents). For $n \geq 6$, $f_{\text{null}}(n)$ rises from 92.0% ($n=6$) to 97.5% ($n=10$), meaning that an overwhelming majority of row/column-preserving $(2,2)$ directions are orthogonal to both diagonals; with order-scaled energy thresholds $s_{\text{fac}}(n) = \max(1, \lfloor (n/4)^2 \rfloor)$, mean restarts decline from 8.03 ($n=5$) to 7.28 ($n=6$), 4.02 ($n=7$), 4.32 ($n=8$), 4.06 ($n=9$), and 3.34 ($n=10$, 258.5 descents).

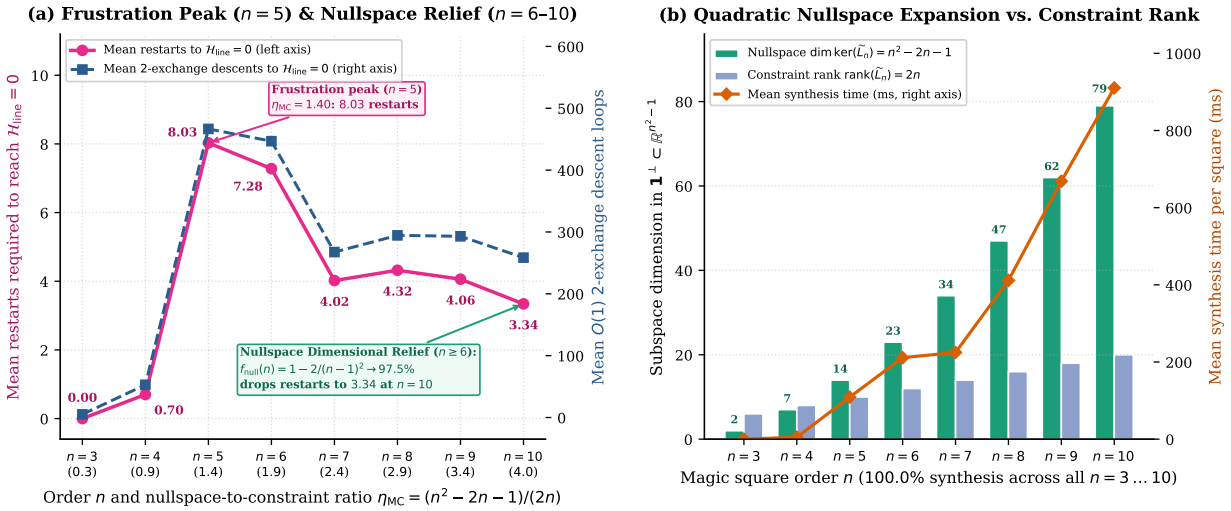


Figure 7: **Multi-order** 100% **ground-state synthesis** ($n = 3 \ldots 10$) **and nullspace dimensional relief.** **(a)** Mean restarts (magenta, left axis) and mean $O(1)$ incremental 2-cycle descent loops (blue dashed, right axis) required by Multi-Exchange Iterated Local Search (ME-ILS) to reach $\mathcal{H}_{\text{line}} = 0$ with 100.0% success across orders $n = 3 \ldots 10$ (Table 8). Combinatorial frustration peaks at $n = 5$ ($\eta_{\text{MC}} = 1.40$: 8.03 restarts, 466.6 descents) and declines across $n = 6 \ldots 10$ (3.34 restarts and 258.5 descents at $n = 10$) as the diagonal-free $(2, 2)$ subspace fraction $f_{\text{null}}(n) = 1 - 2/(n-1)^2$ reaches 97.5%. **(b)** Quadratic expansion of the magic nullspace $\dim \ker(\widetilde{L}_n) = n^2 - 2n - 1$ (green bars, labeled with η_{MC}) versus linear growth of the constraint rank $\text{rank}(\widetilde{L}_n) = 2n$ (blue bars) and mean synthesis runtime per square (orange curve, right axis).

At the same time, because each steepest-descent step evaluates all $\binom{n^2}{2} = O(n^4)$ pair swaps (300 at $n = 5$ vs. 4,950 at $n = 10$), total pair evaluations and wall-clock time grow smoothly with n, reaching $\mathcal{H}_{\text{line}} = 0$ in a mean of 880.8 ms at $n = 10$.

6 Discussion

Across the four papers of the *Magic Gems* series [19, 18, 17], a coherent picture emerges connecting discrete combinatorics, convex geometry, rigid-body mechanics, and permutation energy landscapes. Whereas Papers I–III showed how the magic line-sum constraints $\mathcal{H}_{\text{line}}(S) = 0$ sculpt the convex hull $K(S)$ (Paper I), diagonalize the inertia tensor via bivariate moment orthogonality (Paper II), and stabilize rigid-body rotation and tabletop resting equilibria (Paper III), the present study examines the operator $L_n = A_n^\top A_n$ and the quadratic landscape $\mathcal{H}_{\text{line}}(S) = \mathbf{z}^\top L_n \mathbf{z}$ away from the ground state. Restricting the Rook's-graph-plus-diagonals decomposition of L_n [12, 4, 16] to the zero-mean hyperplane $\mathbf{1}^\perp$ (Proposition 2.1) yields exact integer eigenvalues $\{0, n-2, n\}$ (even n) or $\{0, n-1, n\}$ (odd n) with mean eigenvalue $\bar{\mu}(\widetilde{L}_n) = 2$ and random-permutation expectation $\mathbb{E}[\mathcal{H}_{\text{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4 - 1)$, while linear dipole and diagonal balance ($\langle X, Z\rangle = \langle Y, Z\rangle = R_{D_\pm} = 0$) leaves $2n - 4$ higher-degree polynomial row and column moments unconstrained (Remark 2.2).

Within the magic ground-state manifold $\ker(\widetilde{L}_n)$ itself, our census of sliding-window Shannon entropies $\widetilde{H}_p(k)$ and 2×2 quartet energies (Lemma 3.1 and Proposition 3.2) resolves a second-tier spatial hierarchy. Vanishing interior 2×2 quartet energy is equivalent on all even-order magic squares to vanishing toroidal quartet energy and isolates the 48 Dudeney Group I panmagic squares at $n = 4$, while the Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\text{corner}}(S) - 3\,Q_{\text{edge}}(S)$ explains both the 48 symmetry-protected octupole zeros of Group III ($B_{\text{comp}} = 0$) and the 8 coprime-resonance octupole zeros in Groups IX and X ($B_{\text{comp}} = 8$).

Finally, our exhaustive enumeration of S_9 (Theorem 4.2) and multi-order ablation (Table 7) clarify the roles of continuous versus discrete move neighborhoods on $\mathcal{H}_{\text{line}}(S)$. Naive 2D Gaussian-kernel drift fails (0.0%) under both greedy cell snapping and exact $O(N^3)$ Hungarian bipartite

projection because $\mathcal{H}_{\text{ind}}(t)$ relaxes via convex amplitude averaging in the interior of the assignment polytope; 2-exchange steepest descent traps behind the quadratic self-energy barrier $c_{ab}d^2$ (20.0% at $n = 4$, 1.0% at $n = 5$); and 3-exchange moves break row/column conservation (0/8 on $S^{(3)}_{\text{trap}}$). By contrast, our discrete *Multi-Exchange Iterated Local Search (ME-ILS)* achieves 100.0% ground-state synthesis across all orders $n = 3 \dots 10$ (Table 8) by combining 2-exchange descent, row/column-preserving $(2, 2)$ plaquette moves (which provide the unique escaping $(2, 2)$ move across the $\Delta\mathcal{H}^{\ddagger}_{\text{line}} = +4$ single-swap saddle on $S^{(3)}_{\text{trap}}$, analogous to divergence-free 4-body loop moves in ice-type models [24, 3]), candidate-list $(2, 2)$ and row/column swaps, and threshold-bounded basin-hopping [29]. Across orders $n = 3 \dots 10$, mean restarts peak at $n = 5$ (8.03) and decline to 3.34 at $n = 10$ as the diagonal-free $(2, 2)$ subspace fraction $f_{\text{null}}(n) = 1 - 2/(n - 1)^2$ reaches 97.5%.

Several natural open problems arise from these findings. First, what is the exact number of spurious 2-exchange and line-preserving $(2, 2)$-exchange local minima across the full 16!-state space of S_{16}, and what minimal cycle neighborhood guarantees a 100% single-quench basin on S_{16} as $(2, 2)$ double swaps do on S_9 (Theorem 4.2)? Second, how does the zero-mean spectrum of Proposition 2.1 generalize to three-dimensional magic cubes S_{n^3} [25, 26] and pandiagonal Laplacians? Third, in place of naive 2D spatial kernel interpolation, can a principled concave quadratic-assignment relaxation $\min_{P \in \mathcal{B}_{n^2}} \text{tr}(P^\top L_n P \mathbf{z}\mathbf{z}^\top) - \gamma\|P\|_F^2$ over the Birkhoff polytope $\mathcal{B}_{n^2}$ of doubly stochastic matrices converge directly to permutation vertices in $\ker(\widetilde{L}_n)$ via Sinkhorn entropic annealing? All interactive simulators, verification scripts, and datasets supporting this work are openly available for further exploration.

Data and Code Availability

The interactive browser laboratory (`physicalSolver`), the exact Python verification suite (`verify_paper_d.py`), the certified JSON census (`results_paper_d.json`), and the figure generation pipeline (`make_figures_d.py`) are publicly available in the consolidated repository https://github.com/kylemath/magicGem and live at https://kylemath.github.io/magicGem/web/physicalSolver/.

Competing Interests

The author declares no competing financial or non-financial interests.

Acknowledgments and Declaration of Generative AI Use

This work grew out of the author's interactive physical and computational explorations of magic-square energy landscapes [20], and forms the fourth part of the *Magic Gems* series alongside [19, 18, 17]. Large language models (OpenAI's ChatGPT, Anthropic's Claude, and Google's Gemini) were used by the author as interactive assistants for symbolic checks, verification scripting, interactive simulator prototyping, and manuscript copy-editing following George M. Whitesides's iterative outline-and-figure methodology [31]; all mathematical formulations, proofs, and exact computational certificates were verified by the author, who takes full responsibility for the work.

References

[1] W. S. Andrews. *Magic Squares and Cubes*. Dover Publications, 2nd edition, 1960.

[2] Elwyn R. Berlekamp, John H. Conway, and Richard K. Guy. *Winning Ways for Your Mathematical Plays, Volume 4*. A K Peters, Wellesley, MA, 2nd edition, 2004.

[3] Steven T. Bramwell and Michel J. P. Gingras. Spin ice state in frustrated magnetic pyrochlore materials. *Science*, 294(5546):1495–1501, 2001.

[4] Andries E. Brouwer and Willem H. Haemers. *Spectra of Graphs*. Springer, New York, 2012.

[5] C. R. Calladine. Buckminster Fuller's "tensegrity" structures and clerk Maxwell's rules for the construction of stiff frames. *International Journal of Solids and Structures*, 14(2):161–172, 1978.

[6] Ian Cameron, Adam Rogers, and Peter D. Loly. Signatura of magic and Latin integer squares: isentropic clans and indexing. *Discussiones Mathematicae Probability and Statistics*, 33(1-2):121–149, 2013.

[7] Philippe Codognet and Daniel Diaz. Yet another local search method for constraint solving. In *International Symposium on Stochastic Algorithms: Foundations and Applications (SAGA 2001)*, volume 2264 of *Lecture Notes in Computer Science*, pages 73–90. Springer, 2001.

[8] Simon de La Loubère. *A New Historical Relation of the Kingdom of Siam*. T. Horne, London, 1693. English translation of *Du Royaume de Siam*, Paris, 1691.

[9] Persi Diaconis. Group representations in probability and statistics. *Institute of Mathematical Statistics Lecture Notes—Monograph Series*, 11:1–198, 1988.

[10] Henry E. Dudeney. *Amusements in Mathematics*. Thomas Nelson and Sons, London, 1917.

[11] Bernard Frénicle de Bessy. Des quarrez ou tables magiques. In *Divers ouvrages de mathématique et de physique, par Messieurs de l'Académie Royale des Sciences*. Imprimerie Royale, Paris, 1693.

[12] Chris Godsil and Gordon Royle. *Algebraic Graph Theory*. Springer, New York, 2001.

[13] Kyozi Kawasaki. Diffusion constants near the critical point for time-dependent Ising models. I. *Physical Review*, 145(1):224–230, 1966.

[14] Scott Kirkpatrick, C. Daniel Gelatt, and Mario P. Vecchi. Optimization by simulated annealing. *Science*, 220(4598):671–680, 1983.

[15] Michael J. Lee. Linear algebra and magic squares. *The College Mathematics Journal*, 27(3):198–202, 1996.

[16] Peter D. Loly, Ian Cameron, Walter Trump, and Dan Schindel. Magic square spectra. *Linear Algebra and its Applications*, 430(10):2659–2680, 2009.

[17] Kyle Elliott Mathewson. Balance in inequality: Testing the physical equilibrium, precession, and structural stability of magic gems, 2026. Companion paper III, arXiv preprint (submit/8319388) and submitted to *Journal of Mathematics and the Arts*.

[18] Kyle Elliott Mathewson. Bivariate polynomial moments and Prouhet–Tarry–Escott identities of semi-magic squares, 2026. Companion paper II, arXiv preprint.

[19] Kyle Elliott Mathewson. Magic gems: Convex hulls of lifted magic squares. *arXiv preprint arXiv:2512.09170*, 2026. Companion paper I.

[20] Kyle Elliott Mathewson. `physicalSolver`: Interactive spectral decomposition, multiscale window entropy, and bubble-trinity langevin laboratory for magic squares. https://kylemath.github.io/magicGem/web/physicalSolver/ and https://github.com/kylemath/magicGem, 2026.

[21] James Clerk Maxwell. On the calculation of the equilibrium and stiffness of frames. *Philosophical Magazine*, 27(182):294–299, 1864.

[22] Marc Mézard, Giorgio Parisi, and Miguel Angel Virasoro. *Spin Glass Theory and Beyond*. World Scientific, Singapore, 1987.

[23] Kathleen Ollerenshaw and David S. Brée. *Most-Perfect Pandiagonal Magic Squares: Their Construction and Enumeration*. Institute of Mathematics and its Applications, Southend-on-Sea, 1998.

[24] Linus Pauling. The structure and entropy of ice and of other crystals with some randomness of atomic arrangement. *Journal of the American Chemical Society*, 57(12):2680–2684, 1935.

[25] Adam Rogers and Peter D. Loly. The inertia tensor of a magic cube. *American Journal of Physics*, 72(6):786–789, 2004.

[26] Adam Rogers and Peter D. Loly. The electric multipole expansion for a magic cube. *European Journal of Physics*, 26(5):809–813, 2005.

[27] Pascal Van Hentenryck and Laurent Michel. *Constraint-Based Local Search*. MIT Press, Cambridge, MA, 2005.

[28] David J. Wales. *Energy Landscapes: With Applications to Clusters, Biomolecules and Glasses*. Cambridge University Press, Cambridge, 2003.

[29] David J. Wales and Jonathan P. K. Doye. Global optimization by basin-hopping and the lowest energy structures of Lennard-Jones clusters containing up to 110 atoms. *The Journal of Physical Chemistry A*, 101(28):5111–5116, 1997.

[30] James F. Ward. Vector spaces of magic squares. *Mathematics Magazine*, 53(2):108–111, 1980.

[31] George M. Whitesides. Whitesides' group: Writing a paper. *Advanced Materials*, 16(15):1375–1377, 2004.

A Exact Verification Suite and Reproducibility Manifest

All mathematical identities, spectral decompositions, exhaustive S_9 and order-4 censuses, continuous kernel trajectories, and multi-order local-search benchmarks reported in this paper are certified by the standalone Python verification script `verify_paper_d.py`, which outputs the machine-readable JSON dataset `results_paper_d.json`. First, `verify_section2_spectral()` constructs the $(2n+2) \times n^2$ incidence matrix A_n and operator $L_n = A_n^\top A_n$ for all $3 \le n \le 8$, verifies the irrational eigenvalues $\lambda_\pm(n)$ on $\mathbb{R}^{n^2}$ (Equation 5), projects L_n onto an orthonormal Householder basis of $\mathbf{1}^\perp$ to verify the exact integer spectrum $\mathrm{spec}(\widetilde{L}_n|_{\mathbf{1}^\perp})$ (Equation 6), certifies $\mathrm{tr}(\widetilde{L}_n) = 2(n^2-1)$ and $\mathbb{E}[\mathcal{H}_{\mathrm{line}}(S)] = 2\|\mathbf{z}\|^2 = \frac{2}{3}n^2(n^4-1)$ (Proposition 2.1), and checks all residuals and moments of the 4×4 low-mode counterexample S_{low} (Remark 2.2). Second, `verify_section3_multiscale()` evaluates $\widetilde{H}_p(k)$ ($p \in \{1,2,4\}$, $1 \le k \le n$), interior and toroidal 2×2 quartet energies $W^{\mathrm{int}}_{2\times2}$ and $W^{\mathrm{tor}}_{2\times2}$, complement-pair break counts B_{comp}, the Concentric Shell Quadrupole Identity $M_{xyz}(S) = -8\,Q_{\mathrm{corner}}(S) - 3\,Q_{\mathrm{edge}}(S)$, and mixed octupole moments M_{xyz} across all 880 Frénicle order-4 magic squares (partitioned by Dudeney Groups I–XII) and across constructive families at $n \in \{3,4,5,6,8\}$, certifying Lemma 3.1 and Proposition 3.2. Third, `verify_section4_swap_landscape()` evaluates $\mathcal{H}_{\mathrm{line}}(S)$ and all 36 transposition deltas $\Delta\mathcal{H}_{\mathrm{line}}(\tau_{ab})$ via Lemma 4.1 across all $9! = 362{,}880$ permutations of S_9, verifies that the 8 Lo Shu states and the 8 antipodal traps $D_4 \cdot S^{(3)}_{\mathrm{trap}}$ are the unique 2-cycle local minima in S_9, confirms that all 168 directed 3-cycles strictly increase $\mathcal{H}_{\mathrm{line}}$ on $S^{(3)}_{\mathrm{trap}}$, proves that the row-preserving double swap $(6 \leftrightarrow 7)(3 \leftrightarrow 4)$ is the unique escaping $(2,2)$ move among all 378 disjoint double swaps (crossing the intermediate single-swap saddle at $\mathcal{H}^{\ddagger}_{\mathrm{line}} = 12$ and restoring the steepest-descent basin of S_9 from 270,442 to 362,880/362,880, Theorem 4.2), and runs the 2,000-trial steepest-descent, 3-cycle, and line-preserving $(2,2)$-cycle escape census on S_{16}. Fourth, `verify_section5_bubble_trinity()` integrates the continuous 2D spatial kernel equations (Equation 21) across Regimes I–IV (evaluating both greedy cell-snapping and exact $O(N^3)$ Hungarian minimum-weight bipartite matching), benchmarks $(2+3)$-exchange simulated annealing against multi-exchange local search across budgets $200 \le B \le 2{,}500$, performs the controlled 6-tier operator ablation study (Table 7), and certifies 100.0% multi-order synthesis, per-step pair evaluations, and the nullspace-to-constraint ratio $\eta_{\mathrm{MC}}(n)$ across all orders $n \in \{3,4,5,6,7,8,9,10\}$ for Multi-Exchange Iterated Local Search (Table 8).